IDENTIFYING BUSINESS STATES AND PRODUCT STATES IN TRANSACTION DATA

A Thesis

Presented to the

Faculty of

California State Polytechnic University, Pomona

In Partial Fulfillment

Of the Requirements for the Degree

Master of Science

In

Statistics and Applied Mathematics

By

Arturo Molinar III

2026

COMMITTEE MEMBERSHIP

THESIS: IDENTIFYING BUSINESS STATES AND PRODUCT STATES IN TRANSACTION DATA

AUTHOR: Arturo Molinar III

DATE SUBMITTED: Spring 2026

Department of Mathematics and Statistics

Dr. Manuchehr Aminian
Thesis Committee Chair
Mathematics and Statistics

Dr. Noé Vidales
Mathematics and Statistics

Dr. He Jiang
Mathematics and Statistics

ACKNOWLEDGMENTS

I would like to thank my family for their continuous support. I am grateful to my friends and all those who have helped me throughout my journey. I also thank Dr. Aminian and my committee members for their guidance and mentorship.

ABSTRACT

Daily transaction data reflects both how purchases are distributed across products and how overall demand changes throughout time. These components, discrete, multivariate, and temporal are often analyzed separately, limiting the ability to identify patterns that capture overall business performance.

This study proposes a state-based framework that separates product transaction proportions from demand volume to identify recurring patterns in transaction data over time. Hierarchical clustering is used to identify product preference clusters and further partition these into product trend clusters. Product demand clusters are identified separately based on transaction volume, and product trends are modeled using the Dirichlet distribution to summarize their composition and variability. The proposed framework offers a structured approach to identify business states and product states, providing insight into operational performance, inventory requirements, and purchasing behavior.

Applied to e-commerce data from an electronics store with 2162 products from January 5th 2020 to July 14th 2020, the framework identifies two periods of product preferences, five product trends, and three product demand clusters. By intersecting the product preference clusters with product demand clusters, business states are constructed as an overview of operations based on transactions and product shares. Intersecting product trend clusters with product demand clusters produces product states that provide more detailed insight into purchasing patterns.

We use sequence analysis to examine the temporal behavior of the cluster assignments and resulting states. We give recommendations based on these findings to inform operational decision-making, inventory strategies, and marketing initiatives.

Contents

List of Tables

List of Figures

Chapter 1

Introduction

Understanding business performance requires more than measuring aggregate sales alone. Daily transaction data has temporal, discrete, and multivariate components that, when utilized together, can describe how transactions are distributed between products, how overall transactions fluctuate, and how business performance evolves over time. Analyzing these components simultaneously is essential to identify meaningful insights into general purchasing behavior and operational conditions.

However, existing approaches often analyze these components in isolation. Methods that focus on aggregate demand overlook how transactions are distributed among products, while discrete methods frequently ignore variation in demand intensity or temporal dynamics. As a result, important interactions between product preferences and demand remain difficult to identify.

In this work, we represent each day of transactions as instances of a discrete random variable that records transactions across all products. The rows, dates, are normalized by their respective row margin to estimate the probability distribution for each date. This representation captures the proportions between products while remaining invariant to the overall scale, allowing comparisons of days independent of overall volume. By simultaneously clustering the estimates of the probability distribution of the collective products as well as the variation among the products, we build latent states that reflect business performance and initiatives. By examining the temporal ordering of these states, we are able to further understand and interpret purchasing behavior.

Building on this representation, we propose a state-based framework for analyzing aggregated transaction data. The framework integrates hierarchical clustering, probabilistic modeling, and sequence analysis to jointly capture three key dimensions of business activity:

- product proportions among transactions,
- overall product demand based on transaction volume,
- temporal transitions between clusters and states to identify reoccurring patterns.

These components are used to identify latent states, business states and product states, that describe an overview of operational conditions and purchasing behavior to inform business performance, inventory strategies, and measures for marketing effectiveness.

Business states are found through the overlap between product preference clusters found and product demand clusters for a period of time to understand an overview of business operations. Product states are constructed similarly, except that we use product trends with product demand over a period of time to get a more in-depth interpretation of purchasing behavior.

This framework identifies business states that provide an overview of product preferences and demand, while product states are characterized by more detailed product trends and demand over time.

We now formalize the problem statements considered in this study.

1.1 Problem Statement

Although transaction data has multiple components to analyze, existing approaches lack a unified framework to jointly represent and interpret them to generate data driven recommendations and solutions. In particular, product proportions, transaction volume, and temporal behavior are often analyzed separately, making it difficult to identify recurring patterns that reflect both general operational conditions and changes in purchasing behavior.

A key limitation of current methods is the absence of a structured distinction between high-level summaries of business operations and more detailed patterns in product trends. Without this distinction, it is challenging to simultaneously characterize business states, which capture overall product preferences and demand, and product states, which reflects how product trends and demand evolve over time.

This study addresses these challenges by proposing a framework that separates and relates these complementary perspectives. The objective is to identify latent business states and product states to improve business performance.

More specifically, this thesis investigates the following questions:

- How can daily transaction data be represented to jointly capture the probability distribution the products come from and variation of overall product demand while comparing the transitions over time?
- Can clustering methods identify product preference clusters, product trend clusters, and product demand clusters to identify latent business states and product states?
- How do transitions between these states reveal changes in product behavior and operational conditions over time?

We now discuss the contribution made from the proposed framework and methods used throughout the study.

1.2 Contribution

This thesis makes the following contribution:

1. **State-based representations of Business Performance**

   We introduce a framework that analyzes daily product transactions as temporal, multivariate, and discrete components to identify business states and product states. Both provide context into business performance from operational and marketing perspectives.

Through the use of these methods:

1. **Multi-level clustering of product distributions**

   We apply hierarchical clustering using angular distance to identify clusters of date vectors that are nearly collinear to each other and identify product preferences. We then apply the Jensen–Shannon distance to understand differences between elements of the discrete random variables and identify product trends.

2. **Probabilistic modeling of product trends**

   We use the Dirichlet distribution to model product distributions within product trend clusters, allowing us to approximate the true distributions found by the clusters.

3. **Temporal analysis of state transitions**

   We analyze transitions between business states and product states using sequence-based methods, revealing how operational conditions and product behavior evolve over time.

We now formalize the definitions of the terminology introduced thus far.

1.3 Business States and Product Behavior

Throughout this study, we distinguish between product-level patterns and broader operational conditions. The associated notation is further discussed in Section 2.1. In general, assignment variables are used as placeholders for cluster membership, where m, g, and d denote product preference, product trend, and product demand assignments, respectively. Similarly, b and r denote business state and product state assignments. Throughout this paper, integer-valued assignments are used to denote cluster and state membership.

Product preferences describe groups of dates clustered according to similarity in the direction of their product distribution vectors. They capture similarities between product probability distributions while allowing for variation in relative product proportions due to the underlying geometric interpretation. This representation is well suited for providing an overview of how customers prefer products through their share of transactions.

$$z_m^{(\text{preferences})} = \{x_i \mid \hat{\theta}_i \in C_m\} \tag{1.1}$$

Product trends describe more detailed structure within product preferences by further partitioning existing preference groups into distinct probability distributions. Rather than summarizing overall composition, product trends reflect differences in product transaction shares across days that would otherwise be grouped under similar preference patterns.

$$z_g^{(\text{trends})} = \{x_i \mid \hat{\theta}_i \in C_g\} \tag{1.2}$$

where $C_g \subseteq C_m$.

Product behavior describes temporal changes in product trends. By analyzing transitions between trend assignments, we identify how purchasing patterns change, persist, or shift over time.

$$t : z_t^{(\text{trends})} = i,\ z_{t+1}^{(\text{trends})} = j \tag{1.3}$$

where t indexes the dates and i and j denote possible trend assignments.

Product demand refers to the overall volume of transactions on a given date, independent of product proportions. This captures variation in demand across dates, reflecting differences in total transaction counts regardless of how transactions are distributed among products.

$$z_d^{(\text{demand})} = \{x_i \mid x_i \in C_d\} \tag{1.4}$$

Business states are defined by the joint interaction of product preference clusters and product demand clusters. Two days may share similar product proportions but differ in demand, or exhibit similar demand levels with different product transaction shares. By considering both dimensions together, business states provide a more complete representation of operational conditions.

$$B_b = \{x_i \mid z_i^{(\text{preferences})} = m,\ z_i^{(\text{demand})} = d\} \tag{1.5}$$

Product states are defined by the joint interaction of product trend clusters and product demand clusters. Similar to business states, two days may share similar product trends but differ in demand, or exhibit similar demand levels with different product trends. This allows recurring patterns in purchasing behavior to be identified.

$$P_r = \{x_i \mid z_i^{(\text{trends})} = g,\ z_i^{(\text{demand})} = d\} \tag{1.6}$$

Now that the terminology and corresponding representations have been established, we provide an overview of the approach used throughout the study.

1.4 Overview of Approach

The goal of this analysis is to detect changes in market conditions, inventory needs, and customer behavior by identifying business states and product states.

First, we measure similarity between days using the angular distance as a dissimilarity measure and apply hierarchical clustering to group days with similar product compositions. This step provides an initial partition of the data into product preferences, capturing broad patterns in product transaction shares.

Second, we represent each day as a discrete probability distribution to analyze product transaction shares as estimates to identify dates with similar product preferences. Within each product preferences, we perform sub-clustering using the Jensen–Shannon distance, which directly compares empirical distributions and highlights differences in product composition [1]. This refinement identifies product trends within broader product preferences and can better detect differences in individual product product proportions.

Third, we analyze the overall transaction volume per product as demand by applying a logarithmic transformation to transaction counts and clustering using the Euclidean distance with the Ward linkage. This step captures variation in product demand over days.

To estimate the distribution for product trends, we model the clusters as product distributions using the Dirichlet distribution, providing a probabilistic representation of the typical composition and variability within each trend. Product demand is summarized using descriptive statistics, including measures of central tendency and dispersion. The resulting clusters are used to define latent business states and product states that summarize recurring patterns in both product transaction shares and overall product demand.

Finally, we analyze the temporal sequence of business state assignments and product state assignments using sequence-based methods. This allows for the study of how business states evolve over time and how purchasing behavior shifts across different product trends and demand.

We now discuss additional methods considered for modeling product proportions.

1.5 Other Methods

1.5.1 Hierarchical Dirichlet Process Hidden Markov Model

We originally considered the Hierarchical Dirichlet Process Hidden Markov Model (HDP-HMM), as it performs well for sequential data with an unknown number of latent states. In this framework, latent states are inferred from temporal dependencies in the data, capturing transitions across days while allowing the number of states to grow or shrink as needed [2]. This flexibility makes the HDP-HMM well-suited for uncovering underlying structure in compositional observations through shared statistical strength.

However, the standard HDP-HMM couples state persistence and transition dynamics through a single concentration parameter, which can lead to overly rapid switching between states in practice [2]. To address this limitation, we consider the disentangled (or "sticky") HDP-HMM, which introduces a separate self-transition bias parameter. This modification decouples the tendency to remain in the same state from the overall transition distribution, enabling more realistic modeling of persistent behaviors over time.

In this disentangled formulation, each state is associated with a probability distribution over observations, while transitions between states are governed by a hierarchical Dirichlet process with an added self-transition component [2]. This results in a more interpretable and stable segmentation of temporal patterns, where states correspond to distinct regimes in product trends, and transitions reflect meaningful changes in behavior.

We now discuss the practical limitations associated with the implementation of the HDP-HMM.

Implementation Limitations

In practice, few actively maintained packages exist for implementing the sticky HDP-HMM in common programming languages such as Python and R. Available implementations are often outdated, difficult to adapt, or lack sufficient documentation, which made it challenging to reliably apply the model to our data.

As a result, we instead adopt a hierarchical clustering-based framework that captures similar ideas in a more transparent and implementable way. In particular, our approach mimics key aspects of the HDP-HMM by identifying latent states and analyzing the transitions between them over time.

We now discuss our literature review.

1.6 Literature Review

The paper by van Ruitenbeek et al. (2023) examined how hierarchical agglomerative clustering can improve forecasting accuracy for product sales data with intermittent demand patterns. The authors focused on forecasting challenges caused by fluctuations in demand and frequent zero-sale observations, which make traditional forecasting methods less reliable. Using sales data from more than 3,000 outdoor sports products, the study compared forecasting performance using predefined business product groups versus clustering products based on similarities in sales behavior. The results showed that clustering products using hierarchical agglomerative clustering generally improved forecasting performance compared to traditional business-defined groupings. The authors also found that forecasting performance depended heavily on the characteristics of the time series and that poor aggregation choices could significantly reduce forecasting accuracy. These findings align closely with the current study since both analyses examine how grouping structures and changes in product behavior can impact demand estimation and operational decision-making.

We build on the theory of how hierarchical agglomerative clustering for product sales can detect patterns in time series data for forecasting and related applications [3]. However, rather than using hierarchical clustering strictly for forecasting performance, we use hierarchical methods to infer states between products and aggregated daily transactions while integrating transitions between states and their hierarchical structures. This extends the idea of identifying trends and seasonality through temporal grouping and feature-based grouping [3]. Instead of using Minkowski distance to define time-series similarities, we use alternative distance measures to understand whether products between dates remain relatively proportional to one another. This allows products to be grouped by temporal behavior to better identify dominant seasonal patterns.

Beyond forecasting accuracy, prior work has also examined how behavioral analytics and consumer-focused strategies influence operational decision-making. The nature of our analysis is diagnostic and represents observed behavior over time. Previous studies have highlighted the importance of consumer behavior in informing supply chain forecasting and operational decision-making [4].

The paper by Baldi et al. (2024) examined how consumer-centric supply chain management has evolved in response to changing consumer expectations and retail behaviors. The authors conducted a systematic literature review of 174 articles across 16 leading journals to better understand how consumers influence supply chain management activities and customer experience. The study organized the literature around major supply chain functions and phases of the customer journey while identifying factors that influence consumer-centric supply chain management and resulting consumer outcomes. The authors found that modern supply chains are increasingly shifting toward consumer-focused strategies emphasizing responsiveness, convenience, fulfillment efficiency, and adaptation to rapidly changing customer expectations. Additionally, the study proposed a research agenda encouraging further integration of consumer behavior into supply chain and operational decision-making. Similar concepts are explored in the present study, where behavioral patterns and operational structures are used to understand changes in purchasing activity and product demand over time.

The proposed framework provides insight into common behavioral patterns and transitions, enabling businesses to assess their current state and anticipate future trends. We plot the states as a categorical time series, allowing periods of consistency, seasonality, and transitions between behaviors to be visualized over time. Lastly, we utilize n-grams to identify recurring patterns within the sequence and compare composite sequences of multiple states so that historical sequences may help foreshadow current and future trends.

In this study, purchasing behavior is evaluated through aggregated product-level patterns rather than individual transactions. Specifically, behavior is characterized through product occurrence probabilities and demand intensities, capturing both the relative composition of purchases and overall transaction volume. While these representations provide insight into purchasing dynamics, consumer behavior is also influenced by broader social, psychological, and cultural factors documented throughout the literature.

By modeling both product composition and demand intensity, temporal patterns and transitions in purchasing behavior can be identified. These transitions provide insight into changing customer preferences and operational conditions, which may inform inventory decisions and strategic planning. Consistent with previous work [4], these analyses support the detection of product trends and demand changes, enabling businesses to adapt operational and inventory strategies accordingly.

Lastly, we discuss the impacts of inventory management. The nature of inventory management is to optimize the sourcing, storing, and selling of products [5]. This process affects the success of a business and influences production schedules, supply chain efficiency, customer satisfaction, and cash-flow optimization [5]. By analyzing relationships between products, businesses can forecast inventory needs more efficiently to maintain customer satisfaction while reducing delays in reordering products and materials.

The paper by Kumar et al. (2024) examined how modern business analytics and data-driven decision-making can improve organizational performance and operational efficiency. The authors focused on the growing role of analytics in understanding customer behavior, forecasting demand, and supporting strategic business decisions in dynamic market environments. The study emphasized that businesses increasingly rely on analytical frameworks to identify patterns in transactional and operational data, allowing organizations to respond more effectively to changes in consumer demand and market conditions. The authors found that integrating analytical methods into business operations can improve forecasting accuracy, resource allocation, and long-term planning while helping organizations adapt to uncertainty and changing customer preferences. This supports the framework proposed in the current study, where transactional product relationships and temporal behavioral patterns are used to better understand changes in purchasing activity over time.

By identifying transitions between product states and demand intensities, businesses may gain insight into operational changes that influence inventory allocation, supply chain planning, and customer demand. Through the analysis of temporal product relationships, businesses may better identify periods of changing demand, allowing inventory and operational strategies to adapt accordingly.

Now that we have introduced the problem statement, outlined the framework, and discussed the relevant literature, we move on to the background of the study.

Chapter 2

Background

This chapter introduces the primary methods used in this study. We begin by describing the data representation and probabilistic framework, followed by the clustering methods and distance measures used to identify underlying structure in the data. Finally, we present the sequence-based methods used to analyze temporal behavior.

Clustering methods are used to group observations based on similarities in product distributions and overall product demand by date, forming the basis for identifying business states and product states. The Dirichlet distribution provides a probabilistic model for product proportions within product trend clusters, capturing variation in product proportions across days. Sequence analysis is then used to model transitions between assignments, allowing us to study how product behavior and operational conditions evolve over time.

This approach integrates clustering, probabilistic modeling, and sequence analysis to identify and interpret latent business and product states, as well as changes in their behavior over time.

We now formally outline our notation and data representation used throughout the study.

2.1 Data Representation and Notation

Let n denote the number of observations (days) and p the number of products. The data is represented as a matrix

$$X = (x_{i,j}) \in \mathbb{N}^{n \times p}, \tag{2.1}$$

where $x_i = (x_{i,1}, \ldots, x_{i,p}) \in \mathbb{N}^p$ represents the vector of product counts for day i.

The total number of transactions for observation i is

$$m_{i\cdot} = \sum_{j=1}^{p} x_{i,j}. \tag{2.2}$$

To identify patterns in product proportions, each observation vector is mapped onto the probability simplex by dividing the transaction counts for each product on a given day by the total number of transactions for that day:

$$\hat{\theta}_i = \left(\frac{x_{i1}}{m_{i\cdot}}, \frac{x_{i2}}{m_{i\cdot}}, \ldots, \frac{x_{ip}}{m_{i\cdot}} \right), \quad \hat{\theta}_i \in \Delta^{p-1} \tag{2.3}$$

where the probability simplex is defined as follows:

$$\Delta^{p-1} = \left\{ \theta_i \in \mathbb{R}^p : \theta_{i,j} \geq 0, \ \sum_{j=1}^{p} \theta_{i,j} = 1 \right\}. \tag{2.4}$$

Let K denote the number of clusters observed from the structure of the data, and let

$$z_i \in \{1, \ldots, K\} \tag{2.5}$$

denote the assignment of the cluster for observation i. Define cluster index sets

$$C_k = \{i : z_i = k\}, \quad n_k = |C_k|. \tag{2.6}$$

We now introduce the Dirichlet distribution as a model for product distributions within a given cluster for the product proportions.

2.2 Dirichlet Modeling of Product Distributions

We model each $\hat{\theta}_i$ as a realization from a latent cluster-specific distribution. Each cluster k is associated with a Dirichlet distribution over the simplex Δ^{p-1}, capturing variability in daily product compositions within that cluster.

Given the latent cluster structure identified in the data, observations assigned to cluster k are modeled as

$$\hat{\theta}_i \mid z_i = k \sim \text{Dirichlet}(\alpha^k), \tag{2.7}$$

where

$$\alpha^k = (\alpha_1^k, \ldots, \alpha_p^k), \qquad \alpha_j^k > 0,$$

following the parameterization described in [6].

We parameterize the Dirichlet distribution as

$$\alpha_j^k = \mu_{k,j}\kappa_k, \tag{2.8}$$

where

$$\kappa_k = \sum_{j=1}^{p} \alpha_j^k \tag{2.9}$$

is the concentration parameter, and

$$\mu_k = \frac{\boldsymbol{\alpha}^k}{\sum_{j=1}^{p} \alpha_j^k} \tag{2.10}$$

represents the mean product distribution for cluster k, with $\mu_k \in \Delta^{p-1}$ and $\kappa_k > 0$.

Under this parameterization,

$$\mathbb{E}[\hat{\theta}_i \mid z_i = k] = \mu_k. \tag{2.11}$$

The concentration parameter κ_k controls the variability of the distribution across categories as shown by [6]. Larger values of κ_k correspond to distributions that are more concentrated around the mean μ_k, while smaller values correspond to greater dispersion, allowing observations to place more weight on individual categories.

This parameterization allows each cluster to be interpreted in terms of both its typical product composition and the variability observed across days. In this framework, the mean vector μ_k represents the characteristic product distribution of a cluster, while the concentration parameter κ_k reflects the consistency of product proportions in the clusters.

We now discuss clustering algorithms and their properties.

2.3 Clustering Algorithms

Clustering is used to identify latent groups by comparing observations using a dissimilarity measure and grouping those with similar characteristics [7]. This allows for the detection of an underlying structure that is not directly observable, as observations within the same cluster share similar properties, while differing from those in other clusters.

Let $d(x_i, x_j)$ denote a dissimilarity function between observations. The pairwise dissimilarity matrix is defined as

$$D_{ij} = d(x_i, x_j), \tag{2.12}$$

which serves as the basis for clustering.

We apply hierarchical agglomerative clustering, which constructs clusters through successive merging of observations based on similarity [7].

At each step, the two closest clusters are merged according to a specified linkage criterion. We now discuss the linkage methods used in the study.

2.3.1 Linkage Methods

In this study, two linkage criteria are used.

Complete Linkage:

$$d_{\text{complete}}(C_k, C_l) = \max_{i \in C_k, j \in C_l} d(i,j), \tag{2.13}$$

which produces compact, well-separated clusters by measuring the distance between the two farthest points belonging to different clusters [7].

Ward's Method:

Ward's method minimizes the total within-cluster sum of squares:

$$E = \sum_{k=1}^{K} E_k, \tag{2.14}$$

where

$$E_k = \sum_{i \in C_k} \sum_{j=1}^{p} \left(x_{i,j} - \bar{x}_{k,j}\right)^2. \tag{2.15}$$

Here, K denotes the number of clusters, C_k is the set of observations assigned to cluster k, $x_{i,j}$ represents the number of transactions for product j on date i, and $\bar{x}_k \in \mathbb{R}^p$ is the cluster mean vector with components

$$\bar{x}_{k,j} = \frac{1}{|C_k|} \sum_{i \in C_k} x_{i,j}. \tag{2.16}$$

This approach is sensitive to differences in magnitude, raw counts, and is therefore well suited to identify clusters based on overall demand [7].

We now discuss the dissimilarity measures used throughout our analysis.

2.4 Dissimilarity Measures

Dissimilarity measures are used to compare observations and construct the pairwise dissimilarity matrices used for clustering. These matrices are symmetric and contain distances computed between observations.

Distance measures quantify differences between observations based on their characteristics. In clustering, they are used to compare observations and identify groups with similar patterns. Different distance measures capture different aspects of the data. Distance measures are nonnegative, symmetric, satisfy the triangle inequality, and equal to zero only for identical observations.

The distance measures used in this study satisfy these properties. Angular distance, based on cosine similarity, compares observations based on relative product composition, while Jensen–Shannon distance quantifies differences between product distributions. In contrast, the log-transformed Euclidean distance captures variation in overall transaction volume for products. Together, these measures provide complementary views of differences in product composition, proportions, and demand.

We first discuss measures used to compare product proportions and distributions, followed by a measure used to capture variation in overall transaction volume and magnitude.

2.4.1 Angular Distance

Angular distance measures similarity in relative product proportions by comparing the direction of vectors while remaining invariant to magnitude. As a result, observations with similar compositions but different scales are treated as similar. The angular distance between two vectors is defined as

$$d_{\text{ang}}(x_i, x_j) = \arccos\left(\frac{x_i^\top x_j}{\|x_i\|\|x_j\|}\right). \tag{2.17}$$

This measure is used to cluster observations with similar product proportions [7]. Geometrically, observations can be interpreted as points on a hypersphere restricted to the positive orthant, where dates with similar product proportions lie in close proximity.

This representation provides a high-level view of dates that exhibit similar product preferences. However, angular distance alone does not capture finer differences in the distributions of individual variables. To identify these more detailed distinctions in product transaction shares, an additional distribution-based dissimilarity measure is required.

We now discuss the Kullback-Lieber Divergence to motivate using the Jensen-Shannon Distance as a dissimilarity measure.

2.4.2 Jensen–Shannon Distance

We first introduce the Kullback–Leibler divergence, as it motivates the construction of the Jensen–Shannon distance as a measure of dissimilarity between probability distributions. The Kullback–Leibler divergence is defined as

$$D_{\mathrm{KL}}(\hat{\theta}_i \,\|\, \hat{\theta}_l) = \sum_{j=1}^{p} \hat{\theta}_{i,j} \log \left(\frac{\hat{\theta}_{i,j}}{\hat{\theta}_{l,j}} \right). \tag{2.18}$$

The Kullback–Leibler divergence measures how one probability distribution differs from another by comparing their relative probabilities across product categories. In this setting, it provides a measure of how product distributions differ across observations. However, the Kullback–Leibler divergence is not symmetric:

$$D_{\mathrm{KL}}(\hat{\theta}_i \,\|\, \hat{\theta}_l) \neq D_{\mathrm{KL}}(\hat{\theta}_l \,\|\, \hat{\theta}_i),$$

so the measured difference depends on the ordering of the distributions. This motivates the Jensen–Shannon distance, which extends the Kullback–Leibler divergence to obtain a symmetric and bounded measure between probability distributions.

The Jensen–Shannon distance measures the dissimilarity between probability distributions:

$$d_{\mathrm{JS}}(\hat{\theta}_i, \hat{\theta}_l) = \sqrt{\frac{1}{2} D_{\mathrm{KL}}(\hat{\theta}_i \| m) + \frac{1}{2} D_{\mathrm{KL}}(\hat{\theta}_l \| m)}, \tag{2.19}$$

where

$$m = \frac{1}{2}(\hat{\theta}_i + \hat{\theta}_l) \tag{2.20}$$

is the midpoint distribution between $\hat{\theta}_i$ and $\hat{\theta}_l$.

The Jensen–Shannon distance is symmetric and bounded, which makes it suitable for comparing empirical distributions [1]. Unlike Euclidean-based measures, it directly compares probability distributions and captures relative differences in product proportions rather than absolute magnitudes. By comparing each distribution to a midpoint distribution, it measures how probability mass is allocated across products while remaining symmetric.

Compared to alternative measures, Jensen–Shannon distance emphasizes differences in how probability mass is distributed, allowing for clearer separation between distributions [1]. Since the Jensen–Shannon divergence is bounded by the squared variation distance, it provides a stable measure of dissimilarity between empirical distributions [1]. This makes it well suited for obtaining pairwise distance matrices used for clustering based on similarities between product distributions.

We use the Jensen–Shannon distance to compare product distributions across dates in order to identify periods where product preferences change the most. After obtaining the initial hierarchical clusters, the data are subset according to these clusters and Jensen–Shannon distance matrices are computed separately within each subset. This reduces the complexity of the data and allows for the identification of more refined clusters that capture subtle differences known as product trends.

We now discuss how the Jensen-Shannon distance may also be used to compare cluster means with one another and to new observations.

Comparing Cluster Distributions

To compare clusters, we use the Jensen–Shannon distance between the cluster means. This measures differences between the distributions to assess whether clusters represent distinct product trends and allows new observations to be compared to existing clusters.

We now discuss how Euclidean distance is used to cluster overall product volume.

2.4.3 Euclidean Distance

To analyze product demand through aggregated transaction counts, a logarithmic transformation is applied:

$$y_i = \log(x_i + 1). \tag{2.21}$$

This transformation stabilizes variation in product transaction counts by reducing the influence of large values and reducing skewness. As a result, the transformed observations exhibit more balanced variation across products, leading to clusters that are more spherical in shape and better suited for Euclidean distance-based methods [7].

The Euclidean distance is then computed as

$$d_{\text{Euc}}(y_i, y_j) = \|y_i - y_j\|. \tag{2.22}$$

This distance captures differences in overall demand across observations, reflecting variation in transaction magnitude across products.

We now discuss how to validate the clusters for all dissimilarity measures used.

2.5 Cluster Validation

To determine the appropriate number of clusters, we use the silhouette method proposed by [8]. Although this method does not consider a one-cluster solution, our preliminary analysis in Sec. 3.2 suggests that a single cluster for both product distributions and overall product volume is not plausible due to the structure of the data. We observed a sharp decline in product transactions at the beginning of April, indicating a change in overall product volume. At the same time, many products had transaction counts of zero, suggesting changes in product proportions as well.

We now describe the silhouette method and define the quantities used to compute the silhouette width.

For observation i, let

$$a(i) = \text{average distance within its cluster,} \tag{2.23}$$

and

$$b(i) = \min_{C_l \neq C_i} \text{average distance to other clusters.} \tag{2.24}$$

The silhouette width is defined as

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}, \tag{2.25}$$

where $s(i)$ is bounded between $[-1, 1]$ [8].

Silhouette widths measure how well observations fit within their assigned clusters relative to neighboring clusters. Larger values indicate better cluster separation, while values near zero suggest overlapping clusters. Negative values indicate observations that may be assigned more appropriately to a different cluster.

The overall silhouette index is computed as

$$S = \frac{1}{n} \sum_{i=1}^{n} s(i), \tag{2.26}$$

which measures the overall quality of a clustering solution through within-cluster similarity and between-cluster separation. Larger silhouette index values indicate better defined and more separated clusters.

Having a method to validate the clusters, we can now discuss how to examine their sequential behavior through sequence analysis.

2.6 Sequence Analysis

Sequence-based methods are used to analyze the behavior of cluster assignments and states over time. This allows us to study their temporal behavior and relate changes in cluster assignments to changes in overall product transaction volume and product distributions. To better understand transitions between assignments, we consider transition matrices.

2.6.1 Transition Matrices

Let $\{z_t\}$ denote the sequence of cluster or state assignments over time, where t indexes the dates in the dataset and i and j denote possible assignments. The transition matrix P is defined as

$$P_{ij} \approx \frac{\text{Count}\{t : z_t = i, z_{t+1} = j\}}{\text{Count}\{t : z_t = i\}}. \tag{2.27}$$

This captures the probability of transitioning from assignment i to assignment j [9]. The transition matrix therefore summarizes the relative frequency of future assignments conditioned on the current assignment and allows frequent transitions between assignments to be identified. It should be noted that these probabilities are estimated from the observed data, and future observations may transition to different cluster or state assignments not present within the historical data. Therefore, the transition probabilities can be used as a simple forecast of the next assignment.

We now discuss how longer sequential patterns can be analyzed using N-gram models.

2.6.2 N-gram Models

N-gram models capture higher-order dependencies between assignments by analyzing recurring sequences of cluster or state assignments over time. Let $\{z_t\}$ denote the sequence of assignments over time, where t indexes the dates in the dataset. An N-gram is defined as a sequence of n consecutive assignments,

$$(z_t, z_{t+1}, \ldots, z_{t+n-1}). \tag{2.28}$$

The frequency of an N-gram is computed as

$$\text{Count}(z_t, z_{t+1}, \ldots, z_{t+n-1}), \tag{2.29}$$

which measures how often a sequence of assignments occurs in the data.

N-gram models therefore summarize recurring sequences of assignments and allow longer-term temporal patterns to be identified [10].

In this study, cluster and state assignments are treated as sequences in order to analyze recurring patterns in Business States and Product Behavior. By examining repeated sequences of product demand and product trend assignments, we identify recurring temporal patterns within the data. Thus, with a simple forecast and knowledge of longer-term sequence assignments, insights into future business behavior can be obtained to help inform strategic decision making and policy changes.

With the methods and their intended uses now described, we now evaluate the data using the proposed framework.

Chapter 3

Data and Analysis

In this chapter, we assess the dataset and perform a preliminary analysis using time series and correlation to identify basic patterns. We then apply clustering to identify product preferences, product trends, and product demand, which together inform our analysis of product states and business states.

We now discuss the source of our data as well as steps taken to process it for our analysis.

3.1 Data

The dataset was obtained from Kaggle, eCommerce Purchase History from Electronics Store, where it is described as real-world transaction data that has been redacted from Open CDP [11]. The dataset contains 8 columns and 1,048,575 rows, where each row represents an individual transaction. The variables included in the dataset are described in Table 3.1.

According to the dataset description, there are over two million transactions spanning April 2020 through November 2020. However, the subset of data provided contains fewer observations and includes timestamps outside of this reported range. After data cleaning and preprocessing, the usable date range in this analysis is from January 5, 2020 through

Table 3.1: Description of variables from raw data

Variable	**Description**
Event Time	Date and transaction time
Order ID	Unique identifier for each transaction
Product ID	Unique identifier for each product
Category Code	Description of the product
Brand	Name of the retailer
Price	Product price
User ID	Unique identifier for customers

July 14, 2020. A number of observations contained invalid or inconsistent timestamps (e.g., 1970-01-01), which were removed. As a result, the analysis is conducted on the subset of data available.

Purchasing behavior varies over time and across products, so observations are not identically distributed. For the clustering and Dirichlet modeling stages, each day is treated as an independent observation to capture similarities in product composition and demand. The temporal ordering is retained separately for the sequence analysis, where the transitions between states are examined.

We now discuss which variables we will keep in our analysis and derive further temporal features for our analysis.

3.1.1 Data Engineering

"Category Code" and "Brand" contained 23% and 19% missing values, respectively, and were excluded from the analysis due to inconsistency. Order ID and User ID were also removed, as they do not contribute to the aggregated analysis. As a result, the analysis focuses on

Table 3.2: Finalized variables used in our analysis

Variable	**Description**
Date	Aggregated transaction date (derived from Event Time)
Month	Month of the year
Week of Year	Week number within the year
Day of Week	Day of the week
Products 1–48	Product categories (aggregated counts)
Other Products	Aggregated category for products with expected counts less than ten, containing counts from 2114 products.

Event Time and Product ID.

The Event Time variable was decomposed into Date, Month, Week of Year, and Day of Week. Time-of-day information was not used, as transactions are aggregated at the daily level by product.

To ensure sufficient representation across products, we imposed a threshold on expected counts, requiring each product to have an expected value greater than ten. This threshold was chosen to reduce the influence of sparse categories with low counts, which can introduce instability in distance-based clustering and Dirichlet estimation. Products that did not meet this threshold were grouped into a single "Other" category.

After preprocessing and aggregation, the resulting dataset contains 192 days (rows) and 53 columns, consisting of 49 product categories and 4 temporal variables as shown in Table 3.2.

After preprocessing the transactions by aggregating them by product and date, we visualize the resulting data structure in Figure 3.1. For simplicity, the figure displays only the first four products and the first few dates in the dataset, with products represented as integer values since product names and contextual information are unavailable.

Dates	1	2	3	4
1/1/1970	2291	470	1805	744
1/5/2020	1294	288	1097	418
1/6/2020	1246	252	954	476
1/7/2020	1540	317	1223	583
1/8/2020	1017	188	805	385
1/9/2020	882	187	783	336
1/10/2020	1255	254	1028	409
1/11/2020	1663	269	1367	571
1/12/2020	1501	291	1275	523
1/13/2020	861	205	769	355
1/14/2020	891	170	747	309
1/15/2020	817	180	724	316
1/16/2020	861	171	693	328

Figure 3.1: Processed data table showing transactions aggregated by product and date, where products are columns and dates are row indexes.

It is noted that Excel defaults missing date values to 1/1/1970, which explains the presence of this date within the dataset.

We now discuss limitations associated with the data sourcing and preprocessing.

3.1.2 Limitations

This analysis has a few limitations related to the available data. First, products are represented only by integer IDs, so we do not have information about what the products actually represent or how they relate to each other. Because of this, we rely entirely on patterns in the transaction data to infer relationships between products, rather than using meaningful product characteristics.

The data lacks context about the underlying business. While we know the dataset comes from an e-commerce electronics setting, we do not have information about the specific company, customer base, or operational structure. This makes it harder to interpret why certain product behaviors and business states appear and limits how specific or actionable our conclusions can be.

Our choice of aggregation also affects our analysis, interpretation of results, and recommendations. Because we limit the analysis to daily aggregation, we could be missing individual customer behavior as well as purchasing patterns that could be found by aggregating by different time intervals, such as hourly, weekly and monthly.

Finally, the dataset reflects a subset of the original data and contains inconsistencies in timestamp information. Although these issues were addressed through preprocessing, they may still influence the observed patterns and should be considered when interpreting the results.

We now discuss how COVID-19 impacted countries and therefore our data .

3.1.3 Impacts of COVID-19

Given the time frame of the data (January 5, 2020 through July 14, 2020), it is important to consider the significant impact that COVID-19 had on both supply chain operations and employment levels. Although the dataset country of origin is unknown, the effects observed in the United States provide a useful general model for understanding how the pandemic disrupted economic activity globally.

On March 11, 2020, the World Health Organization (WHO) declared COVID-19 a global pandemic. Two days later, on March 13, 2020, the United States declared a national emergency, and by mid-March many states had begun implementing shutdown measures. Shortly thereafter, on March 28, 2020, the Cybersecurity and Infrastructure Security Agency (CISA) updated its guidance on essential critical infrastructure workers, influencing state and local decisions regarding workforce restrictions.

These policy changes occurred alongside major disruptions in global trade and labor markets. The United Nations Conference on Trade and Development (UNCTAD) reports that approximately 80% of globally traded goods are transported by maritime shipping [12]. During the initial months of the pandemic, the United States experienced nearly a 20% increase in canceled maritime shipments [13], while unemployment rose sharply, reaching approximately 13% during the study period [14].

Taken together, these findings suggest that disruptions to maritime shipping and labor shortages associated with social distancing policies contributed substantially to inventory and distribution challenges during the COVID-19 pandemic. Reduced operational capacity likely compounded delays in production and transportation, while decreased product availability and limited labor supply contributed to a broader economic disruption.

With the data source, preprocessing steps, and relevant global events during the time period now discussed, we are ready to perform a preliminary analysis to determine what assumptions may reasonably be made about the data.

3.2 Preliminary Analysis

Before performing the main analysis, we examine the data to understand the underlying relationships between time and product transactions. This step provides context for the patterns present in the data and helps guide the modeling approach used in subsequent sections.

We begin our preliminary analysis by evaluating the time series of the products used in our analysis.

3.2.1 Time series of Products

We begin by evaluating the time series of aggregated product counts to understand how transaction activity evolves over time across different products.

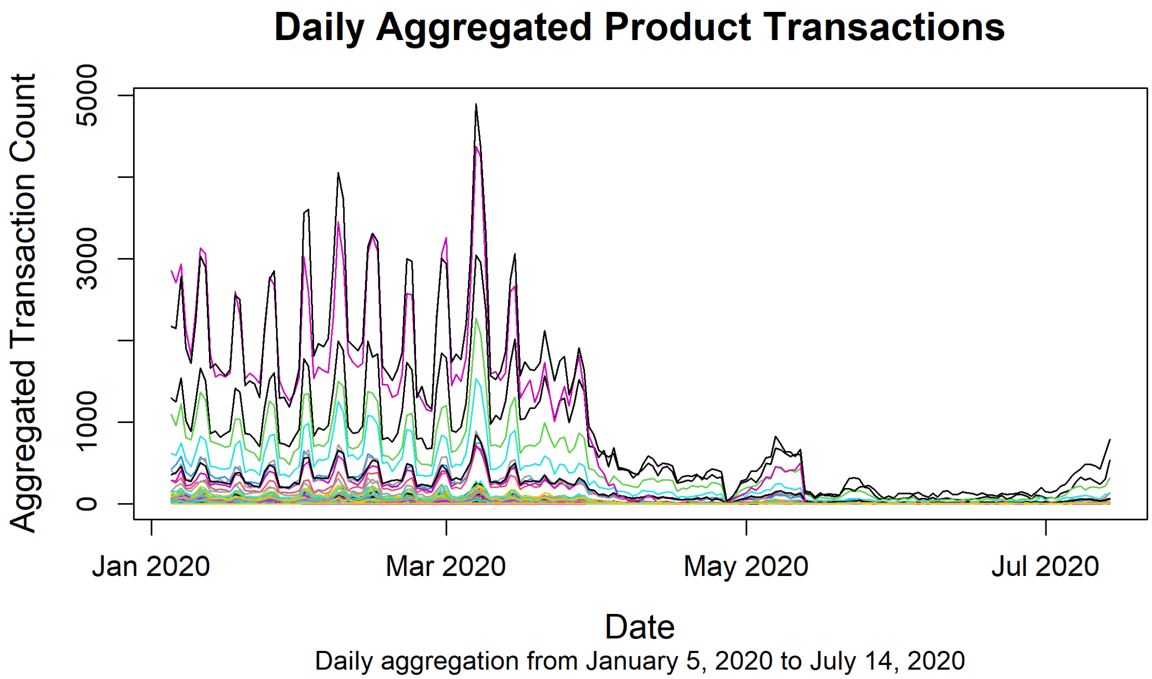


Figure 3.2: Time series of aggregated transactions by Products after preprocessing throughout the duration of the study as dates

As shown in Figure 3.2, product counts generally follow similar patterns over time, suggesting the presence of a common temporal trend across products. There is a noticeable increase in activity in early March, followed by a sharp decline beginning in April.

From May through July, product counts remain low across most categories. This pattern is consistent with broader disruptions during this period, although the data alone do not identify the specific cause. By mid-July, product counts begin to increase again, indicating a partial recovery in overall transactions.

These observations indicate that variation in overall demand affects many products simultaneously, making it difficult to distinguish differences in product composition based only on raw counts.

We now examine the pairwise correlations between the products.

3.2.2 Product Correlations

To further examine the relationships between products, we analyze the pairwise correlations between aggregated product counts.

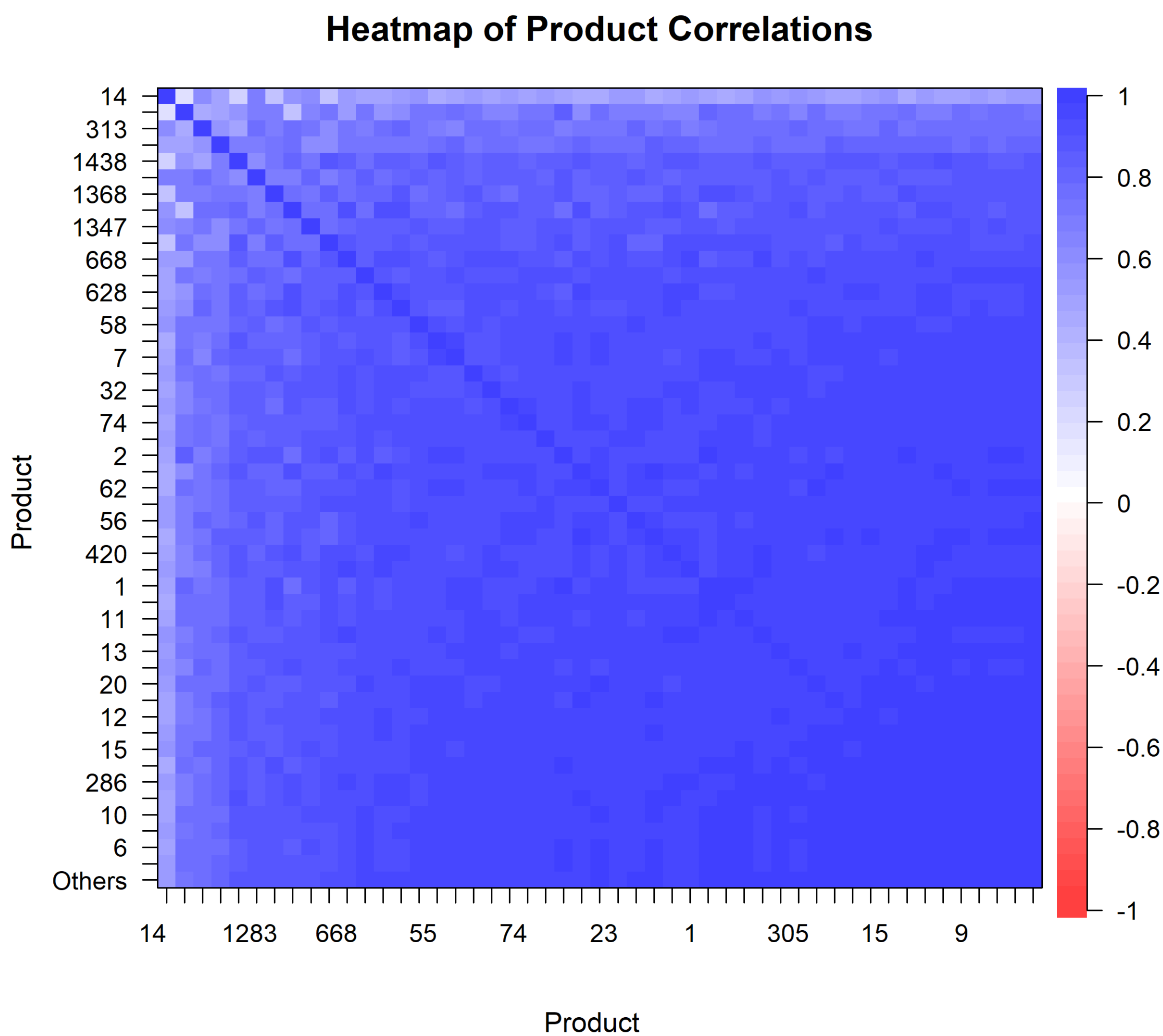


Figure 3.3: Heatmap showing pairwise correlations between products using daily aggregated transaction counts

Evaluating the products without explicitly accounting for time in Figure 3.3, we observe that most of the products are highly correlated with each other, with some showing weaker associations. This pattern is consistent with the shared temporal trends observed in the time series, where increases and decreases in activity affect many products simultaneously.

As a result, these correlations primarily reflect common time-dependent variation rather than direct relationships between individual products. Although some differences between products are present, correlation analysis alone does not isolate variation in relative product composition.

We now summarize our findings from the preliminary analysis.

Preliminary Findings

Together, the time series and correlation analysis show that product activity is influenced by both overall demand and differences in relative composition. However, these effects are not easily separated in the raw data, as shared temporal variation drives much of the observed structure.

These observations motivate the use of clustering methods to separate these components. In the following sections, we identify product states based on relative composition, refine these into product trends, and analyze demand separately to capture variation in overall activity.

Now that we have an understanding of the underlying structure of the data, we can start our analysis of the clusters starting with product preferences.

3.3 Product Preferences

In this section, we identify product preferences based on similarities in relative product composition. We apply hierarchical clustering using angular distance to group days with a similar proportional structure and interpret the resulting clusters. We now begin our analysis using hierarchical clustering on the angular distances to uncover the product preferences.

3.3.1 Angular Distance Hierarchical Clustering

To compare observations, we compute the angular distance between daily product vectors. To aid in interpretation, distances are normalized by dividing by π, mapping values to the interval $[0, 1]$. The resulting dissimilarity matrix is visualized below as a heatmap.

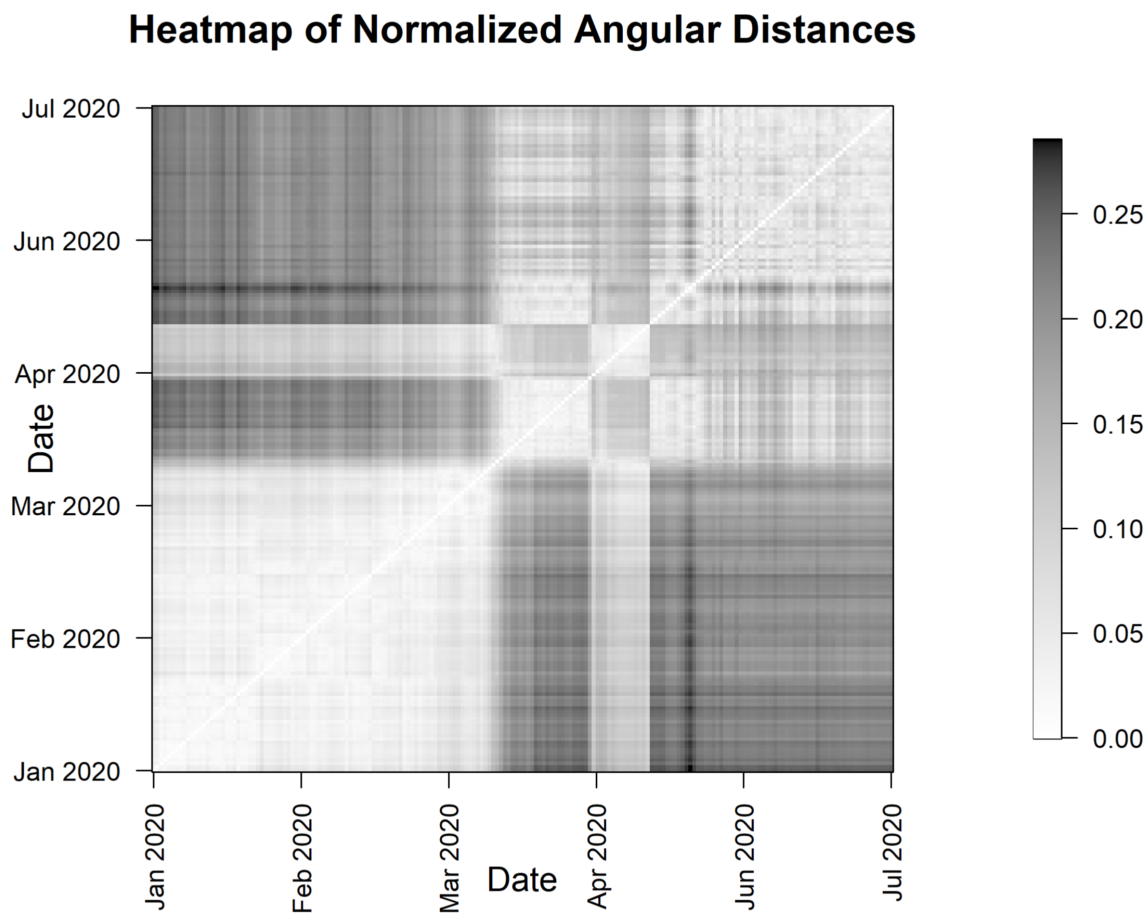


Figure 3.4: Heatmap of normalized angular distances

As shown in the heatmap in Figure 3.4, distinct block structures are observed in the normalized angular distance matrix. From January through the end of March, pairwise distances are relatively small (below 0.05), indicating that these dates share a similar proportional structure across products. From April through July, distances are again relatively small within this period, suggesting a different but internally consistent product profile.

Within this second block, a distinct substructure appears from early May to mid-May, where the distances deviate from the surrounding period. This indicates a temporary shift in the shares in product transactions before returning to the broader pattern. Overall, these patterns suggest the presence of distinct clusters, which we use to guide the clustering analysis.

We next apply hierarchical clustering to the normalized angular distance matrix using the complete linkage criterion, which favors compact and well-separated clusters. To determine the appropriate number of clusters, we examine the average silhouette width as the number of clusters increases.

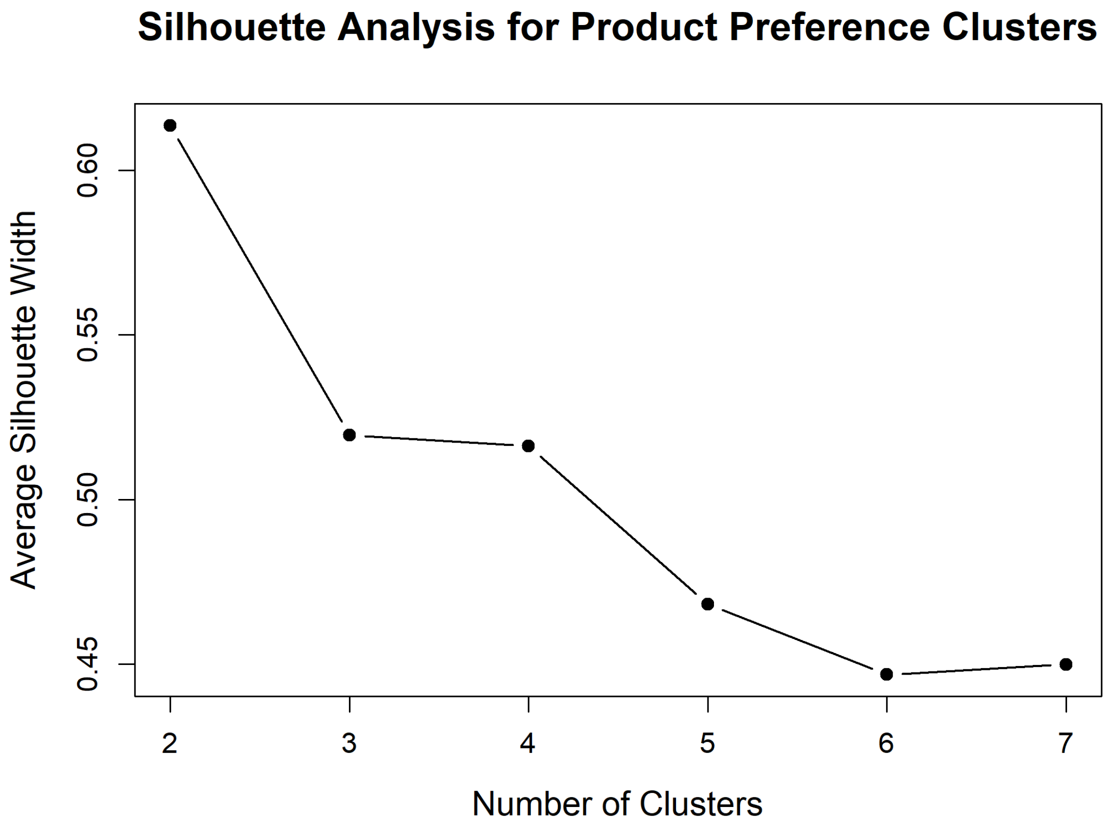


Figure 3.5: Average silhouette width by number of clusters using angular distance for product preference clusters

The maximum average silhouette width occurs at two clusters in Figure 3.5. The next highest value is observed at three clusters, followed by four clusters with similar performance. This suggests that while two clusters provide the strongest separation, alternative solutions with additional clusters may also capture a finer structure.

To further assess these possibilities, we examine the individual silhouette scores, shown below along with the corresponding cluster assignments.

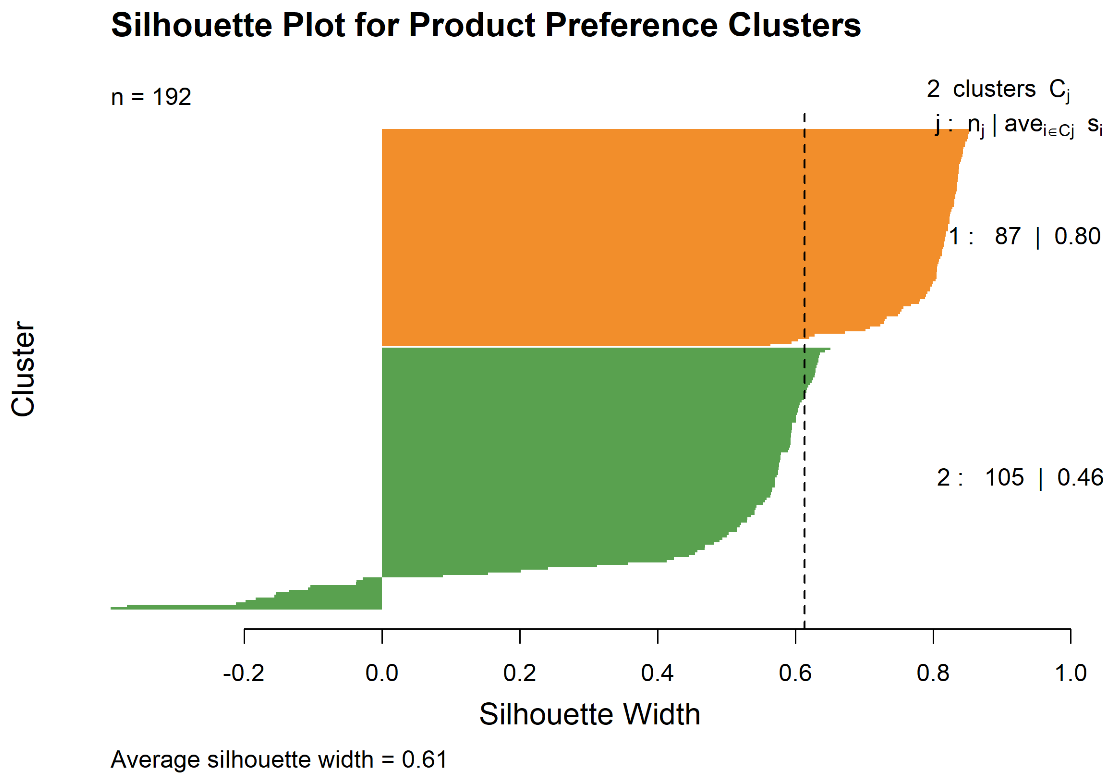


Figure 3.6: Silhouette widths using angular distance for product preference clusters

The two-cluster solution yields an average silhouette width of approximately 0.61, indicating reasonably well-separated groups as shown in Figure 3.6. In contrast, the three- and four-cluster solutions contain a larger proportion of observations with negative silhouette widths, suggesting weaker separation and greater ambiguity in cluster assignments.

Although additional clusters may capture finer variations, the two-cluster solution provides a more stable and interpretable partition of the data. Given the objective of identifying broad patterns in product proportions, we proceed with two clusters.

We next examine the dendrogram to further understand the clustering structure.

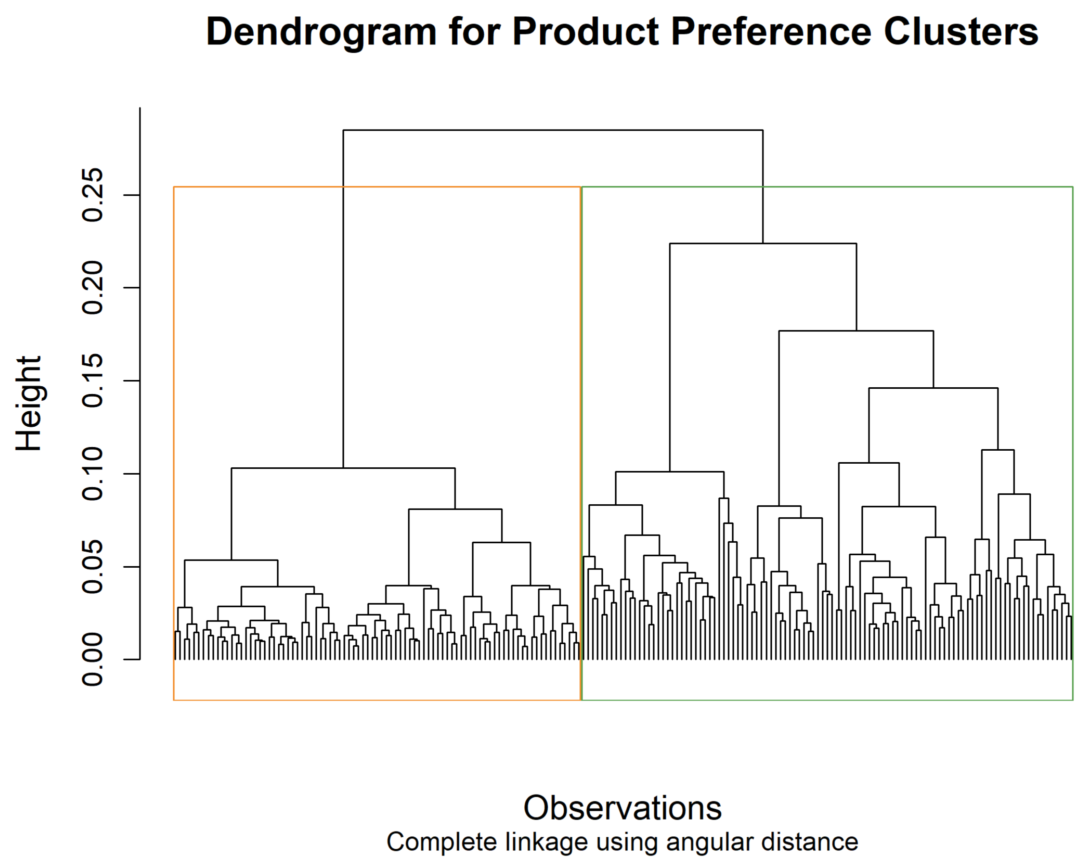


Figure 3.7: Dendrogram of product preferences using complete linkage and angular distance cut at two clusters

From the dendrogram in Figure 3.7, the primary split occurs at a height of approximately 0.25, separating the observations into two main groups. Within each group, additional substructure is present, indicating potential finer structure in the data.

Based on the clustering results and silhouette analysis, we focus on the two-cluster solution, which provides a stable and interpretable partition of the data.

We now analyze the clusters found.

3.3.2 Product Preferences Found

We now examine the product preferences obtained from the clustering using angular distance to understand how they evolve over time.

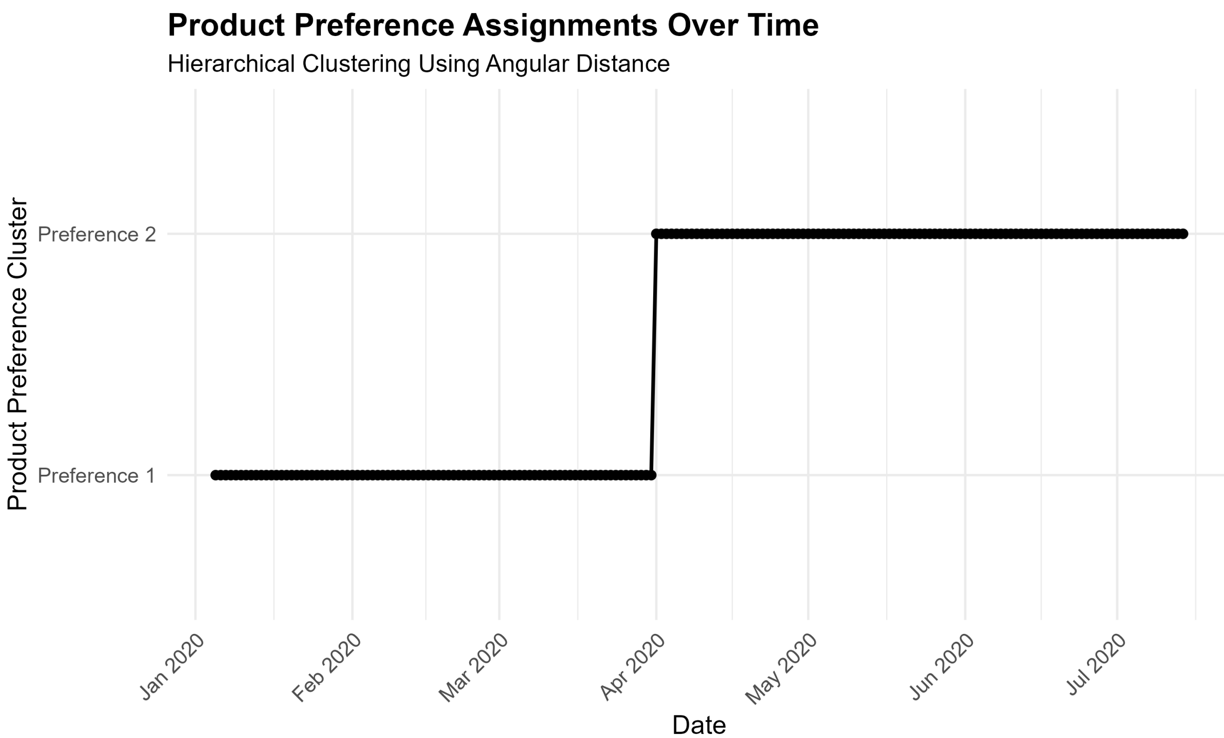


Figure 3.8: Categorical time series of product preference clusters found

From Figure 3.8, we observe that from January through the end of March, product preference cluster 1 remains dominant and relatively consistent. Beginning in early April, there is a clear transition to product preference cluster 2, which persists through mid-July.

This transition aligns with the shift observed in Figure 3.2, suggesting that the two clusters capture distinct periods of product composition. While the timing is consistent with broader external disruptions, the clustering results themselves primarily reflect changes in relative product distributions rather than total demand.

We summarize the product preferences as follows.

Table 3.3: Product preference clusters summary

Preference Clusters	Count	Proportion (%)	Months Occurred
1	87	45	Jan–Mar
2	105	55	Apr–Jul

Table 3.3 summarizes two primary product preference clusters corresponding to different product transaction shares over time. These clusters provide a structured representation of changes in overall product proportions throughout the observed period.

While product preferences capture broad differences in the vector composition of dates, they do not capture finer variation within these groups at the individual product level. We therefore refine these clusters to identify product trends.

3.4 Product Trends

In this section, we refine the product states by identifying variation in product composition within each state. To capture differences in product distributions, we apply hierarchical clustering using the Jensen–Shannon distance.

3.4.1 Jensen-Shannon Distance Hierarchical Clustering

We examine the Jensen–Shannon distance matrices as heatmaps to identify product trends within each product preference clusters.

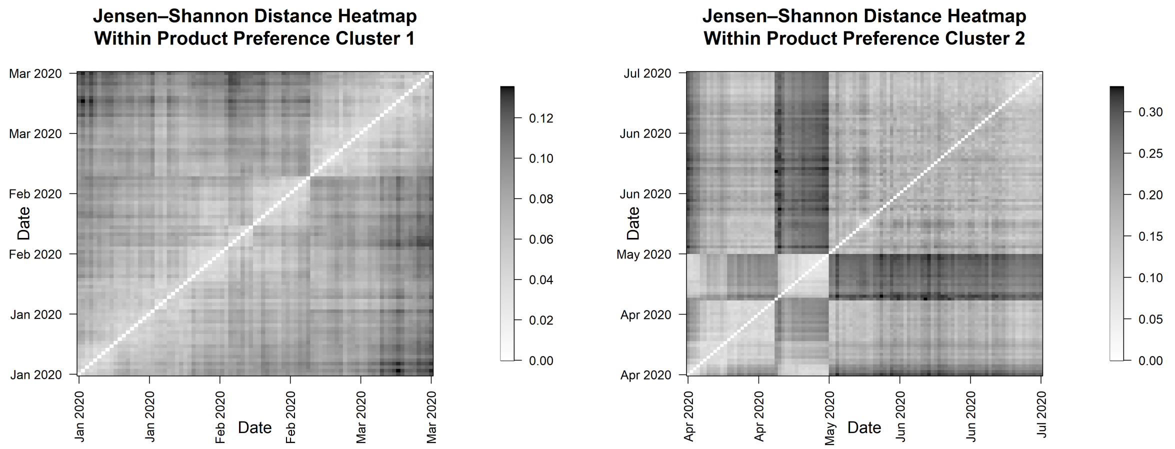


Figure 3.9: Jensen–Shannon distance heatmaps for product preference clusters 1 and 2

From the heatmaps in Figure 3.9, we observe different structural patterns across the two product preference clusters. For product preference cluster 1, there are no clearly defined block structures as observed with angular distance. Instead, there is a gradual progression of dissimilarity from January through March, suggesting continuous transitions in product proportions over time that were not displayed before.

In contrast, product preference cluster 2 shows clear block structures, indicating the presence of more distinct subgroups. This suggests that product proportions within cluster 2 are more distinct compared to those in cluster 1.

These differences in structure suggest that product trends vary in both continuity and separation between clusters, motivating additional clustering of product preferences.

We apply hierarchical clustering using the complete linkage criterion and examine the average silhouette width as the number of clusters increases.

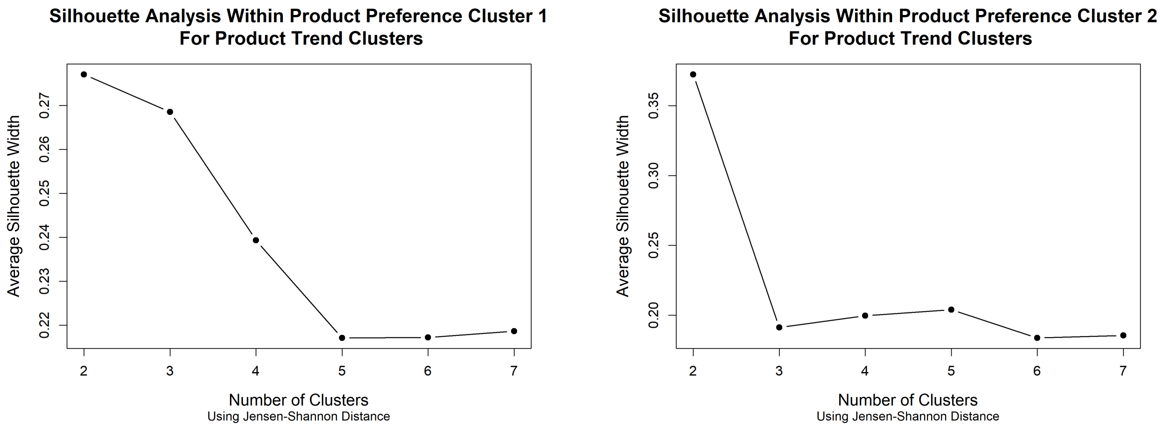
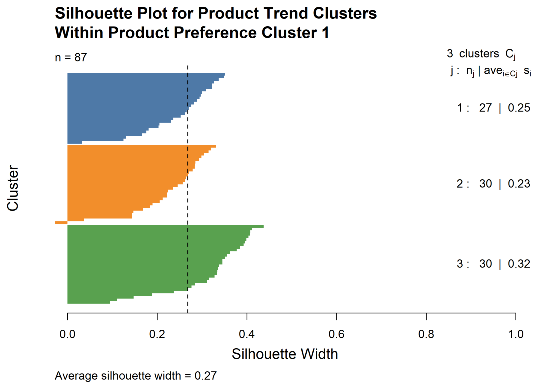


Figure 3.10: Average silhouette width by number of subclusters using Jensen–Shannon distance for product trends within product preference clusters 1 and 2

For product preference cluster 1, the silhouette plot in Figure 3.10 suggests that both two- and three-cluster solutions are reasonable. Given the gradual structure observed in the heatmap from Figure 3.9, we select the three-cluster solution to better capture this variation.

For product preference cluster 2, the silhouette plot in Figure 3.10 shows a clear maximum at two clusters, with significantly lower values for higher cluster counts. This indicates that a two-cluster solution provides the most stable partition.

We further assess cluster quality by examining the silhouette widths for each observation.



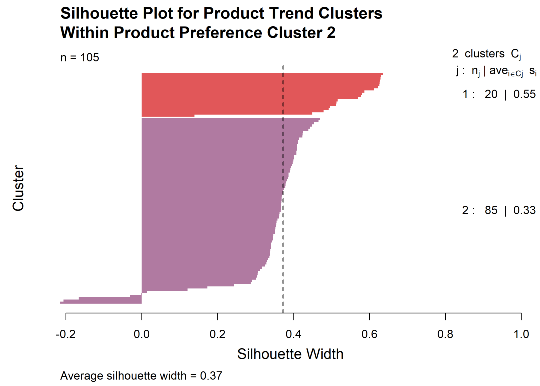


Figure 3.11: Average silhouette width by number of subclusters using Jensen–Shannon distance for product trends within product preference clusters 1 and 2

As shown in Figure 3.11, the three-subcluster solution for product preference cluster 1 yields an average silhouette width of approximately 0.27, indicating moderate cluster separation. This solution captures multiple product trends while maintaining acceptable cluster quality.

For product preference cluster 2, the two-subcluster solution yields an average silhouette width of approximately 0.37, also indicating moderate separation. Although some overlap between subclusters remains, alternative clustering solutions do not improve overall cluster quality.

We next examine the dendrograms to interpret the clustering structure.

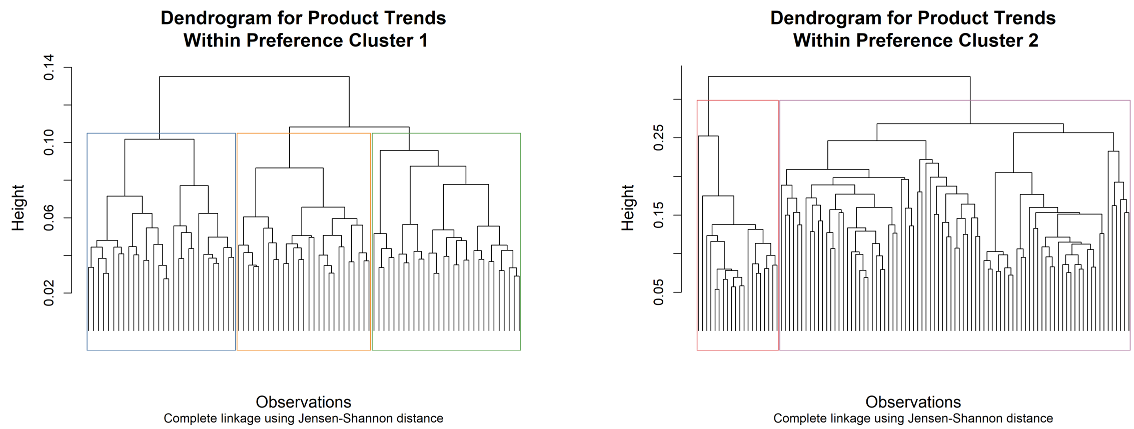


Figure 3.12: Dendrograms of product trend subclusters obtained using complete linkage and Jensen–Shannon distance, cut at three clusters for preference cluster 1 and two clusters for preference cluster 2.

As shown in Figure 3.12, the dendrograms reveal different clustering structures across the two preference clusters. For product preference cluster 1, the primary split occurs at a height of approximately 0.105, resulting in three substructures. This aligns with the gradual transitions observed in the corresponding heatmap.

For product preference cluster 2, the primary split occurs at a height of approximately 0.3, resulting in two substructures. Although this structure is less visually apparent in the heatmap, it is supported by the clustering results and highlights the importance of empirical validation.

Overall, these findings suggest that preference cluster 1 exhibits more gradual variation in product trends, whereas preference cluster 2 contains more clearly defined and separable subgroups.

We now summarize the product trends found.

Summary of Product Trends

We visualize the product trends identified from the second layer of clustering on the product proportions to examine how these trends evolve over time.

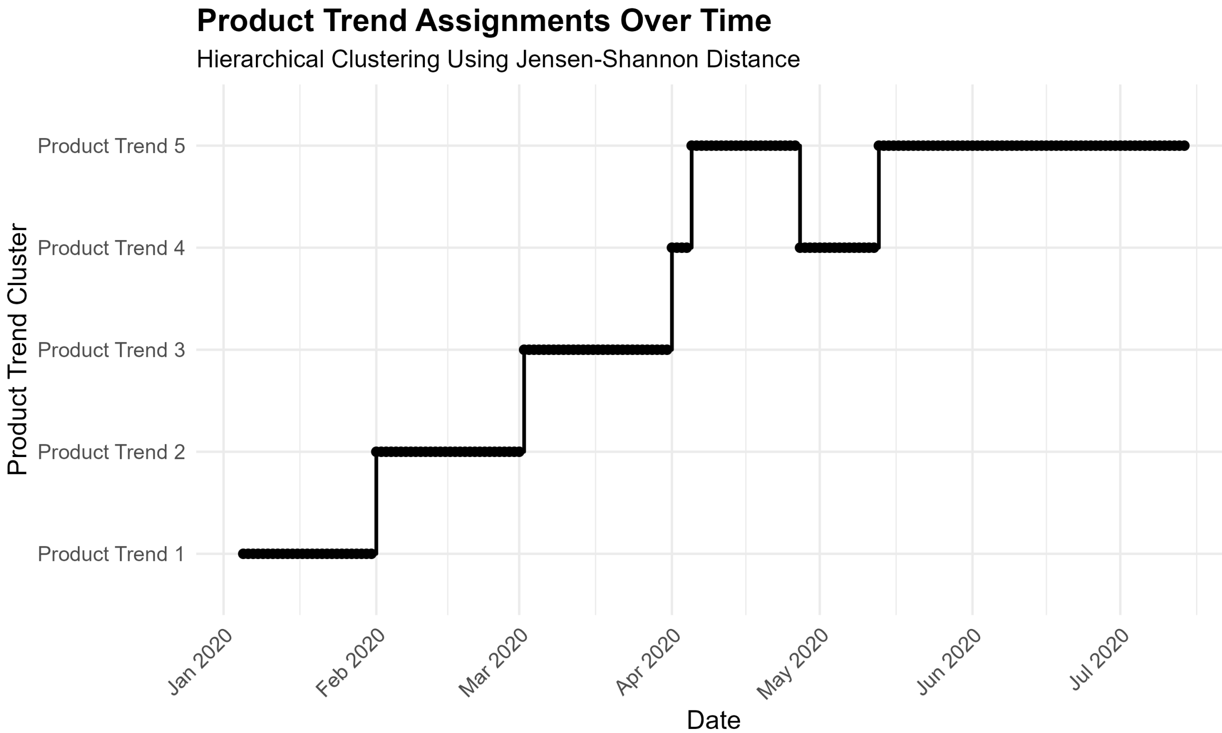


Figure 3.13: Categorical time series of product trends identified from hierarchical clustering of product proportions using Jensen–Shannon distance.

As shown in Figure 3.13, five distinct product trends are observed over time. The first three trends correspond primarily to January through March, with each month largely associated with a different trend.

In April, Trend 4 appears briefly before transitioning to Trend 5 for the remainder of the month. Toward the end of April and into mid-May, the series transitions back to Trend 4. Following this period, Trend 5 becomes dominant and remains prevalent through mid-July.

We summarize the identified product trends in Table 3.4.

Table 3.4: Product trend clusters summary

Product Trends	Count	Proportion (%)	Months Occurred
1	27	14.1	Jan
2	30	15.6	Feb
3	30	15.6	Mar
4	18	9.4	Apr–May
5	85	44.3	Apr–Jul

Now that we have identified and summarized the product trend clusters, we are able to model them using the Dirichlet distribution.

3.4.2 Product Trends Modeled as Dirichlet Distributions

With product trends identified, we model each cluster using a Dirichlet distribution to quantify its compositional structure and variability. The model is fit using the `DirichletReg` package in `R`, assuming a single Dirichlet distribution for all observations within each cluster. The fitted model is then used to estimate the corresponding Dirichlet parameter vector α^k for each cluster.

Summary of Distributions

We analyze the behavior of the Dirichlet concentration parameters and overall concentration levels for the fitted product trends. Table 3.5 summarizes the distribution of α parameters across threshold ranges for each trend.

We analyze the behavior of the Dirichlet concentration parameters and overall concentration levels for the fitted product trends. Table 3.5 summarizes the distribution of α parameters across threshold ranges for each trend.

Table 3.5: Distribution of Dirichlet concentration parameters across threshold ranges for each product trend.

Trend	$\boldsymbol{\alpha < 1}$	$\boldsymbol{1 \leq \alpha < 2}$	$\boldsymbol{2 \leq \alpha < 10}$	$\boldsymbol{\alpha \geq 10}$	$\boldsymbol{\kappa_k}$
Trend 1	0 (0.0%)	0 (0.0%)	24 (49.0%)	25 (51.0%)	4736
Trend 2	0 (0.0%)	0 (0.0%)	25 (51.0%)	24 (49.0%)	3852
Trend 3	0 (0.0%)	0 (0.0%)	27 (55.1%)	22 (44.9%)	3878
Trend 4	0 (0.0%)	0 (0.0%)	39 (79.6%)	10 (20.4%)	1028
Trend 5	21 (42.9%)	10 (20.4%)	16 (32.7%)	2 (4.1%)	367

As shown in Table 3.5, Trends 1–4 appear to be highly concentrated. All of their Dirichlet parameters are at least 2, with a substantial proportion exceeding 10 (ranging from 20.4% to 51.0%), indicating relatively low variability and distributions that are tightly centered around their means [6]. The large values of $\kappa_k = \sum_j \alpha_j$, particularly for Trends 1–3, further support this interpretation by reflecting stable and well-defined compositions.

Trend 5 differs substantially from the other states. A sizable portion of its parameters falls below 1 (42.9%) and between 1 and 2 (20.4%), indicating increased sparsity and greater variability. Its much smaller total concentration ($\kappa_5 = 367$) suggests a more dispersed distribution, where observations are less tightly clustered around a single composition. Overall, trend 5 is more heterogeneous and less stable compared to the other states.

To assess how distinct these distributions are, we next compute pairwise Jensen-Shannon distances between the cluster means.

Validation of Distributions

We compute the pairwise Jensen–Shannon distances between the mean distributions of the five trends.

Table 3.6: Pairwise Jensen–Shannon distances between product trend mean distributions

	Trend 1	**Trend 2**	**Trend 3**	**Trend 4**	**Trend 5**
Trend 1	0.0000	0.0629	0.0840	0.1682	0.3827
Trend 2	0.0629	0.0000	0.0845	0.1589	0.3690
Trend 3	0.0840	0.0845	0.0000	0.1167	0.3379
Trend 4	0.1682	0.1589	0.1167	0.0000	0.2556
Trend 5	0.3827	0.3690	0.3379	0.2556	0.0000

As shown in Table 3.6, Trends 1–3 exhibit relatively small pairwise Jensen–Shannon distances, indicating similar product distributions. Trend 4 shows larger distances from Trends 1–3, reflecting greater differences in product composition. Trend 5 has the largest distances from all other trends, indicating a clear separation from the remaining distributions.

Overall, Trends 1–3 are similar, Trend 4 exhibits moderate differences, and Trend 5 is the most distinct. Comparing trends by their product distributions provides additional information on the separation between them.

3.4.3 Comparing Trends

We compare the identified trends to examine differences in their structure by grouping the mean product proportions from the fitted distributions.

As shown in Table 3.7, Trends 1–4 exhibit similar structures, with only a small number of dominant products and most products contributing between 0% and 5% of the overall proportions. In contrast, Trend 5 displays a noticeable shift, with more products having zero contribution and fewer dominant products. This suggests increased sparsity and concentration, where a smaller subset of products drives overall activity.

We now assess the product behavior throughout the transitions between product trends

Table 3.7: Distribution of mean product proportions within each product trend

Trend	**0%**	$(0, 5]\%$	$> 5\%$
Trend 1	0	44	5
Trend 2	0	44	5
Trend 3	0	44	5
Trend 4	0	44	5
Trend 5	9	36	4

to further our understanding of how the trends relate to one another.

Product Behavior

Product behavior describes how individual products, or products collectively, change over time. From the identified product trends, we can directly examine differences in product preferences across trends. For selected products, Table 3.8 highlights representative behavioral patterns. The complete list of product mean proportions is provided in the appendix for reference.

As shown in Table 3.8, product proportions vary across trends, with some products declining, increasing, or remaining relatively consistent over time. Across Trends 1–3, changes are comparatively gradual, with shifts in product shares but no major structural differences. For example, Product 286 is dominant in Trend 1 and steadily declines across subsequent trends, whereas Products 1 and 3 increase in relative share across the later trends.

The most notable change occurs in Trend 5, where several products decrease to zero contribution and the remaining products account for a larger share of transactions. This reflects a shift toward a more concentrated distribution, where a smaller number of products account for a larger share of transactions while many products contribute little or no share,

Table 3.8: Mean product proportions across trends for selected products

Product	Trend 1	Trend 2	Trend 3	Trend 4	Trend 5	Behavior
286	20.6%	18.7%	17.7%	10.8%	0.4%	Decline
Others	19.7%	20.9%	20.5%	20.2%	27.3%	Increase
1	10.9%	11.0%	13.9%	18.3%	21.1%	Increase
3	8.6%	7.9%	9.0%	11.8%	14.1%	Increase
5	5.5%	6.1%	5.8%	6.3%	7.0%	Increase
308	1.7%	1.5%	0.6%	0.6%	0.0%	Decline
305	1.2%	1.2%	1.0%	0.9%	0.0%	Decline
62	0.3%	0.3%	0.3%	0.3%	0.4%	Consistent

resulting in a less uniform distribution of product proportions.

Given the progression of product shares toward zero, and considering that the business operates in an electronics retail setting where products are generally in demand, this behavior may reflect inventory-related constraints rather than changes in customer preferences. Since product trends capture variation in relative proportions between products but do not account for differences in overall transaction volume, we analyze demand separately to capture variation in magnitude and provide additional context for the observed product behavior.

We now transition from analyzing the product proportions to the aggregated product counts to understand the product demand clusters within the dataset.

3.5 Product Demand

In this section, we identify product demand clusters by analyzing variation in overall transaction volume for products. We apply hierarchical clustering using Euclidean distance to capture differences in demand magnitude and interpret the resulting structure.

3.5.1 Euclidean Distance Hierarchical Clustering

After applying a logarithmic transformation to the raw counts, we compute the Euclidean dissimilarity matrix. Hierarchical clustering is then performed using the Ward linkage criterion, which groups observations based on similarity in magnitude. The resulting dissimilarity matrix is visualized in the heatmap below.

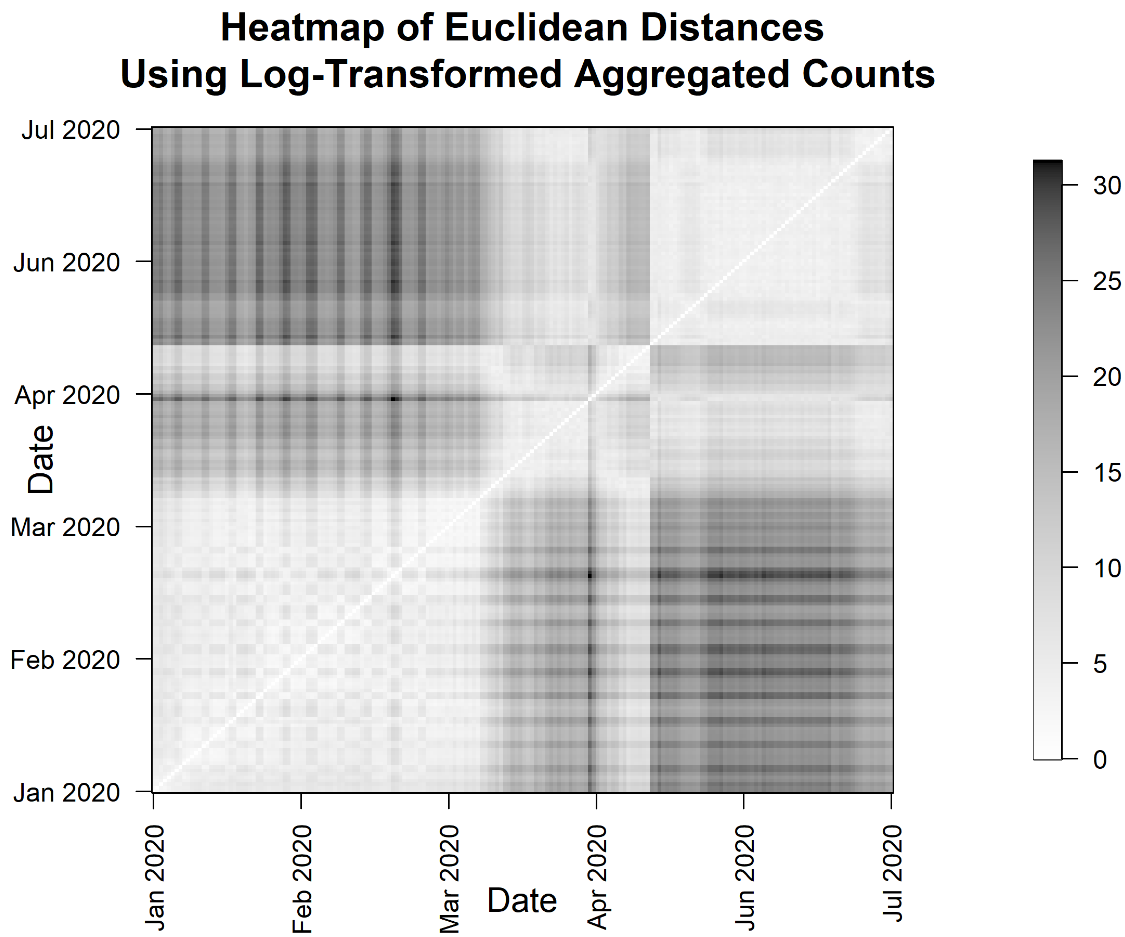


Figure 3.14: Heatmap of Euclidean distances computed from log-transformed aggregated product counts

As shown in Figure 3.14, the heatmap suggests the presence of two, and possibly three, demand clusters. January through March exhibit relatively similar levels of product transaction volume, followed by a gradual decline extending through mid-July. A distinct shift is observed from early May to mid-May, indicating a temporary change in demand before the continuation of the broader downward trend.

To determine the number of clusters, we examine the average silhouette width.

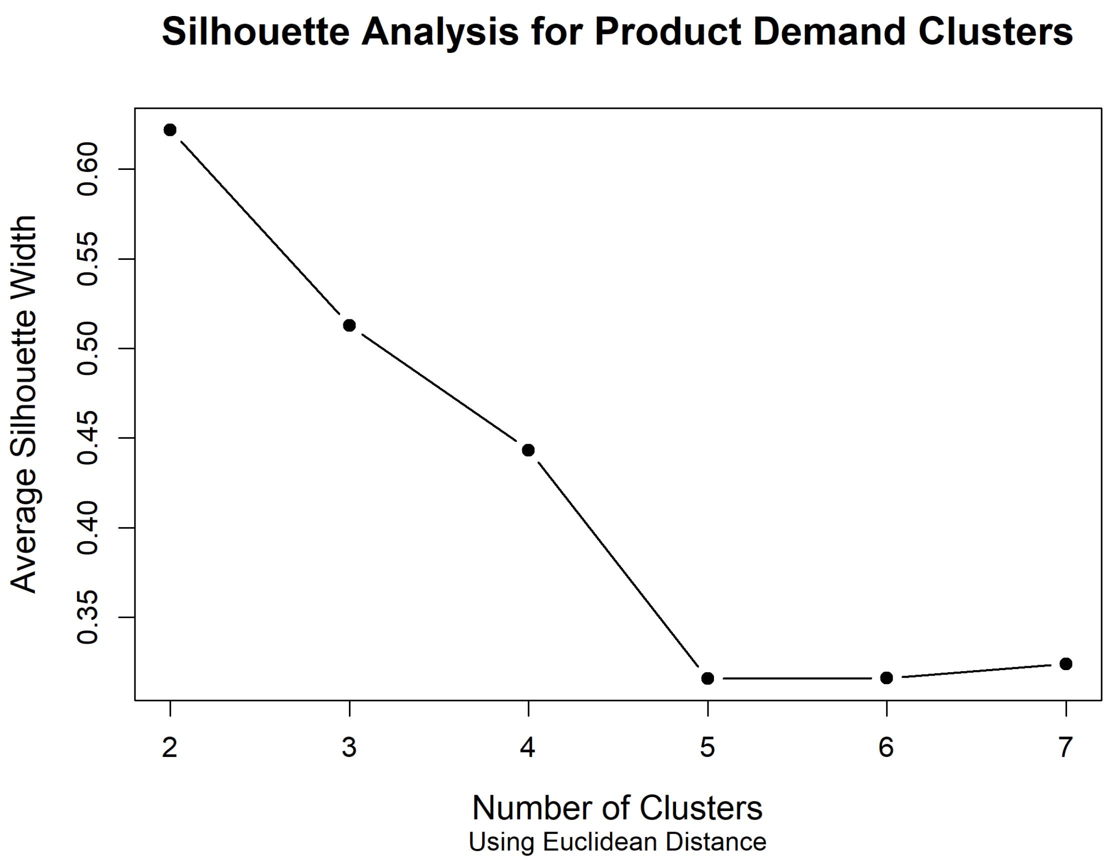


Figure 3.15: Average silhouette width by number of clusters using Euclidean distance for product demand clusters

As shown in Figure 3.15, the maximum average silhouette width occurs at two clusters, with the next highest value observed at three clusters. This suggests that the two-cluster solution provides stronger overall cluster separation.

However, the heatmap in Figure 3.14 indicates the possible presence of three distinct regions. To further evaluate this structure, we examine the individual silhouette widths for the selected clustering solution.

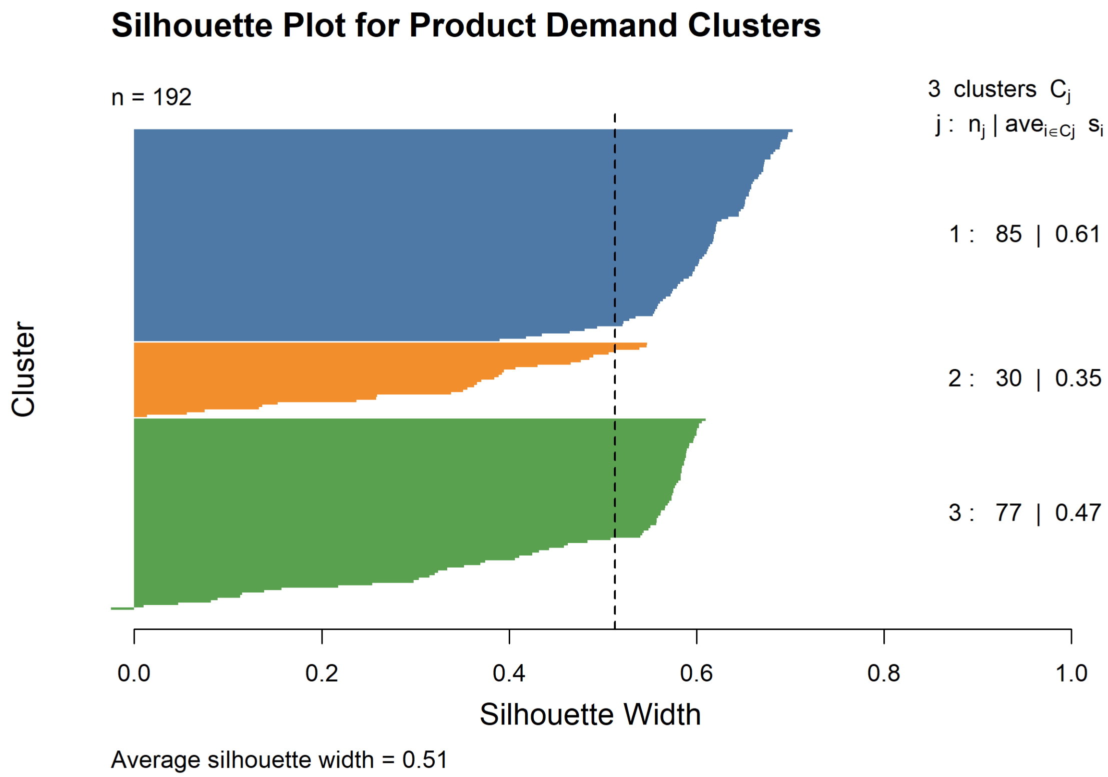


Figure 3.16: Silhouette widths using Euclidean distance for product demand clusters.

As shown in Figure 3.16, the two-cluster solution contains several negative silhouette values, indicating poor assignment for some observations. In contrast, the three-cluster solution exhibits more consistent silhouette widths, suggesting improved separation across groups.

We next examine the dendrogram to interpret the clustering structure.

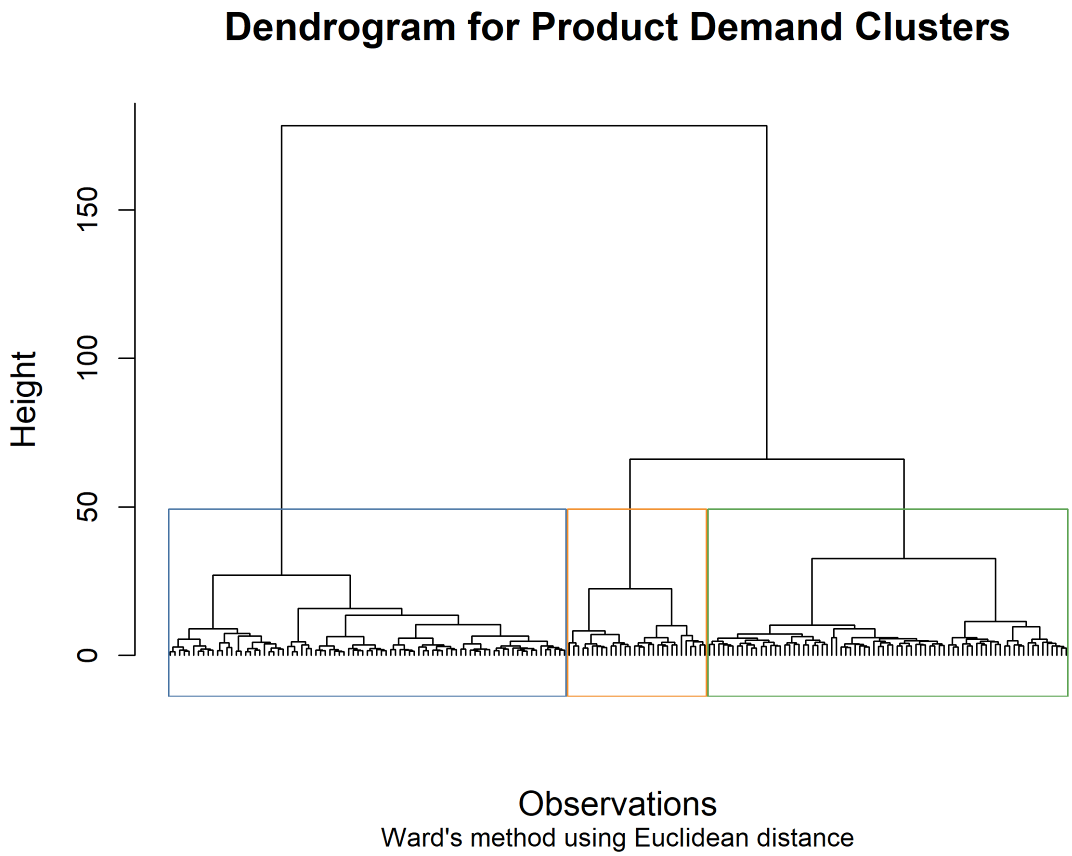


Figure 3.17: Dendrogram of product demand clusters using Ward's method and Euclidean distance cut at three clusters.

As shown in Figure 3.17, the primary split in the dendrogram produces three distinct groups, which is consistent with the structure observed in the heatmap. Based on this evidence, we proceed with the three-cluster solution.

Now that we have found the product demand clusters, we can summarize them.

3.5.2 Summary of Product Demand Clusters

We now examine the product demand clusters to understand how they evolve over time.

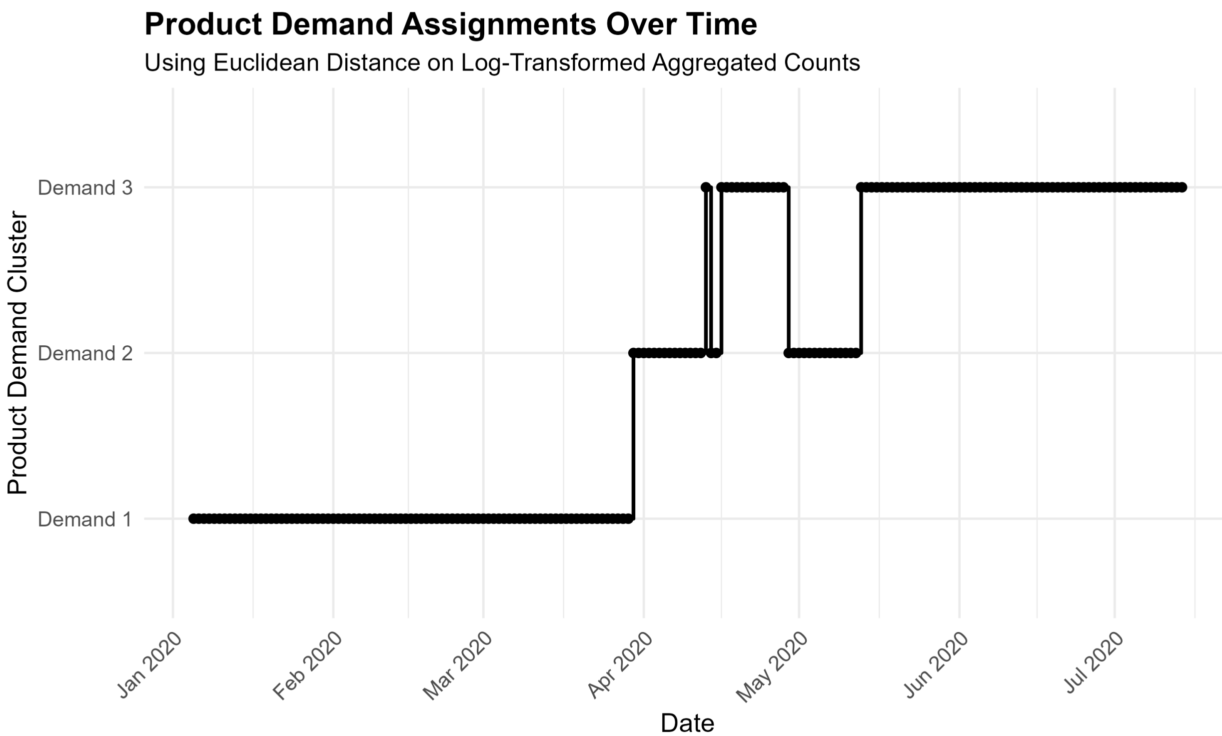


Figure 3.18: Categorical time series of product demand clusters found

As shown in Figure 3.18, product demand remains relatively stable from January through March. In April, there is a transition from product demand cluster 1 to product demand cluster 2, followed by alternating transitions between clusters 2 and 3. Over time, cluster 3 becomes increasingly persistent, indicating a sustained reduction in demand.

Overall, cluster 1 represents high and stable demand, while clusters 2 and 3 capture the transition and subsequent decline in transaction volume. The largest changes occur during April and May, corresponding to the period where demand shifts most rapidly.

This timing aligns with COVID-19–related policy changes that affected inventory shipments and labor availability. While demand appears to persist, and the observed product behavior suggests continued customer preference, the changes may reflect inventory shortage related to reduced workforce capacity, policy constraints, and interruptions in shipments, which limited the volume of products sold.

We summarize the identified product demand clusters in Table 3.9.

Table 3.9: Product demand clusters summary

Product Demand	Count	Proportion (%)	Months Occurred
1	85	44.3	Jan–Mar
2	30	15.6	Apr–May
3	77	40.1	Apr–Jul

We now summarize the distributions of total transactions per day using descriptive statistical methods.

Summary of Product Demand Distributions

To further interpret the demand clusters, we summarize the total number of transactions per day using descriptive statistics. Since clustering is based on similarity in product volume, the row sums provide a measure of the overall demand for each observation within a cluster.

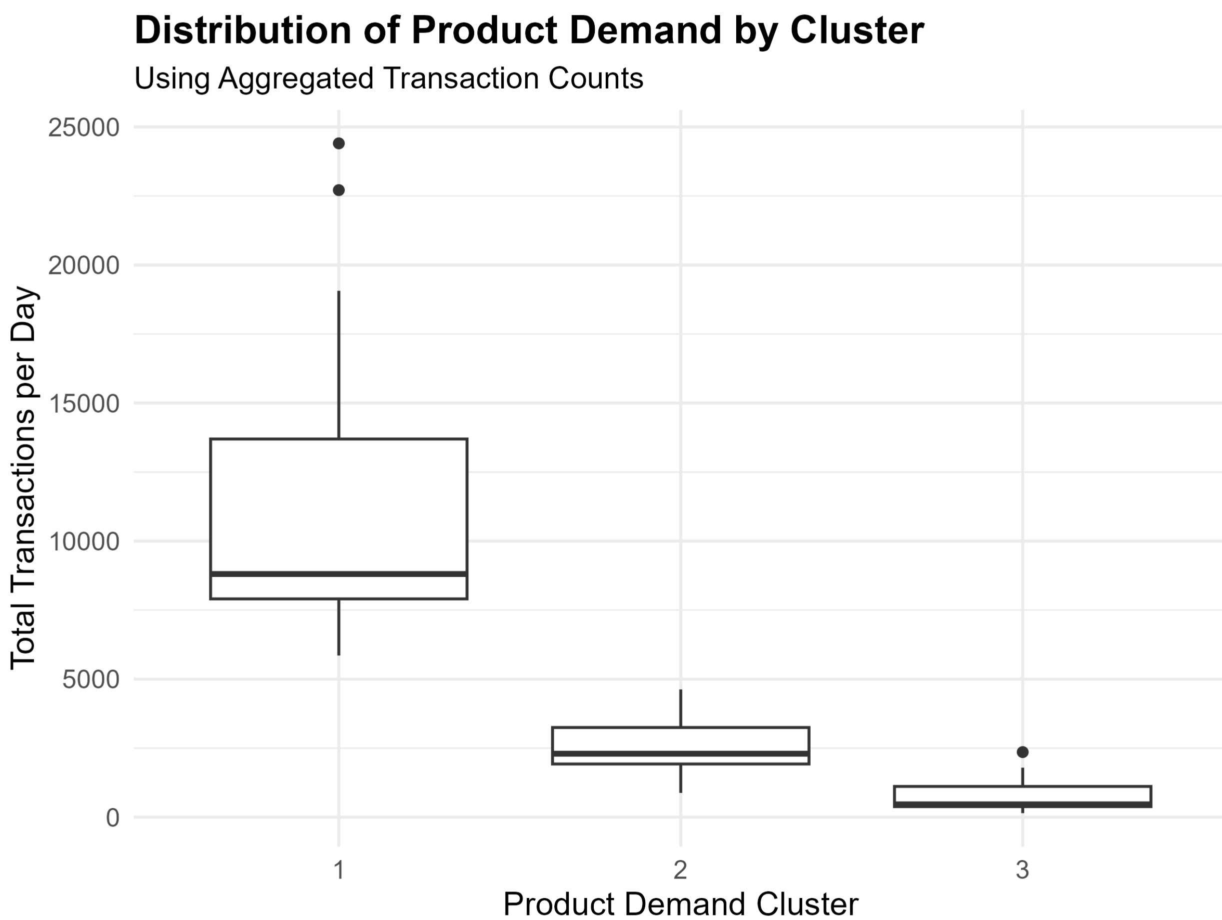


Figure 3.19: Box-plots of total daily transactions across product demand clusters

As shown in Figure 3.19, each product demand cluster exhibits distinct levels of transaction volume with relatively little overlap between their distributions. Product demand cluster 1 corresponds to high transaction volume, product demand cluster 2 corresponds to moderate demand, and product demand cluster 3 corresponds to low demand. This separation indicates that the clustering effectively captures differences in overall product demand levels.

Having identified product preferences, product trends, and product demand clusters, we now examine how these components interact to describe business states and product behavior over time.

Chapter 4

Results

Following the analysis, we construct the business and product states. We begin with the business states and interpret the results.

4.1 Business States

Business states are defined by the joint assignment of observations to the product preference and product demand clusters. A change in either component results in a transition between states, providing a representation of operational performance based on both product proportions and demand.

We visualize the overlap in cluster assignments between product preference and product demand clusters using a categorical time series.

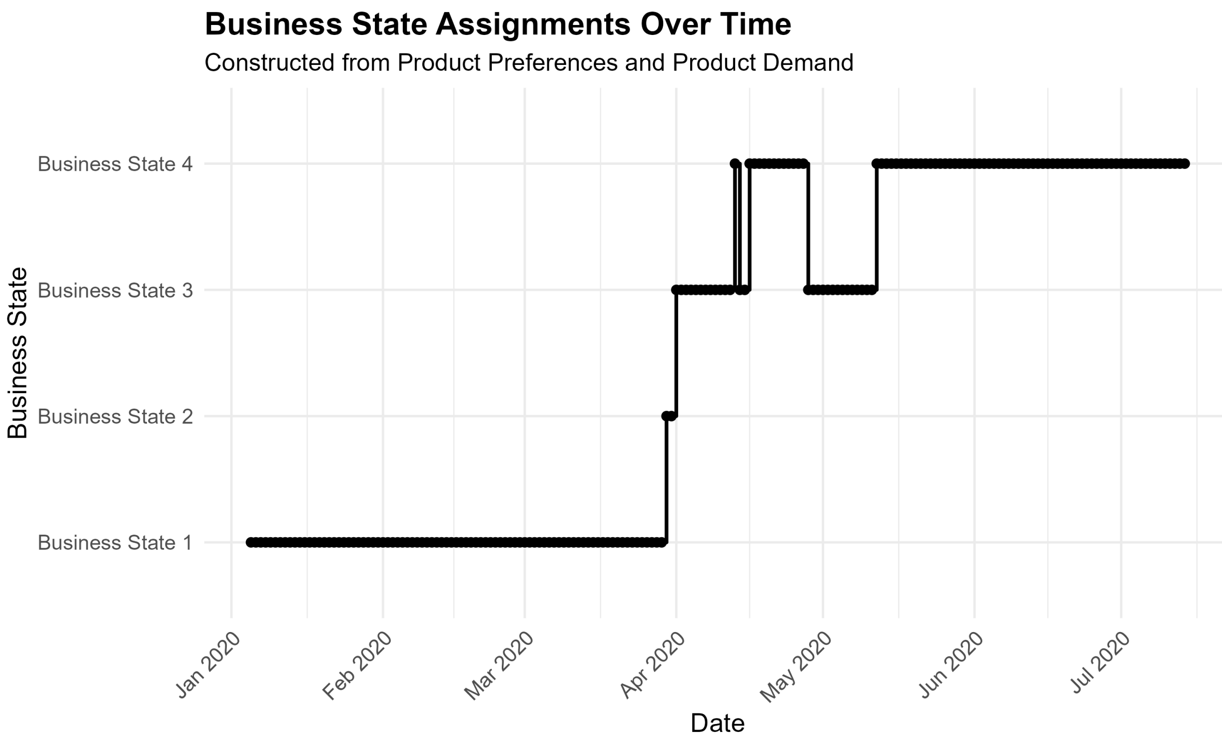


Figure 4.1: Categorical time series of business states identified from the joint assignment of product preference and product demand clusters.

As shown in Figure 4.1, Business State 1 is dominant from January through March, indicating relatively consistent operational behavior prior to the observed disruptions. At the end of March, a brief transition to Business State 2 occurs, characterized by changes in demand while product preferences remain largely unchanged. This suggests that variation in transaction volume, rather than shifts in product preferences, primarily drove the transition.

At the beginning of April, a transition is observed that reflect changes in product preferences, indicating a shift in the relative composition of products. The following months, transitions between states are primarily driven by fluctuations in demand, while product preferences remain relatively stable. This pattern suggests that customer purchasing behavior remains consistent, with variation arising from changes in overall transaction volume.

We summarize the identified business states in Table 4.1.

Table 4.1: Business states summary

Business State	Count	Proportion (%)	Months Occurred
1	85	44.3	Jan–Mar
2	2	1.0	Mar
3	28	14.6	Apr–May
4	77	40.1	Apr–Jul

These observations are consistent with supply-side disruptions, where constraints such as inventory availability or operational capacity affect transaction volume without substantially altering product preferences. The timing of these transitions aligns with COVID-19–related policy changes, which may have contributed to the observed variation in business states.

We summarize the observed transitions and their corresponding behaviors in Table 4.2.

Table 4.2: Summary of business state transitions.

Business State Transition	States Involved	Behavior
Transition 1	State 1 → State 2	Decrease in demand
Transition 2	State 2 → State 3	Change in product preferences
Transition 3	State 3 → State 4	Decrease in demand
Transition 4	State 4 → State 3	Increase in demand

As shown in Table 4.2, changes in product preferences and demand do not occur simultaneously, but instead appear independently across transitions. This further supports the interpretation that demand fluctuations and product composition capture distinct aspects of operational behavior.

We do not construct a transition matrix or apply n-gram analysis due to the limited number of transitions and the absence of frequent or recurring state sequences.

Having established an overview of product demand and product preferences, we now examine product states to capture more detailed variation in product behavior.

4.2 Product States

From the product trends and product demand clusters, we construct the product states through the overlap of their cluster assignments. A change in either component results in a transition between product states. This provides a more detailed interpretation of customer purchasing behavior by jointly capturing changes in product trends and demand.

We now visualize our findings of the product states found as a categorical time series.

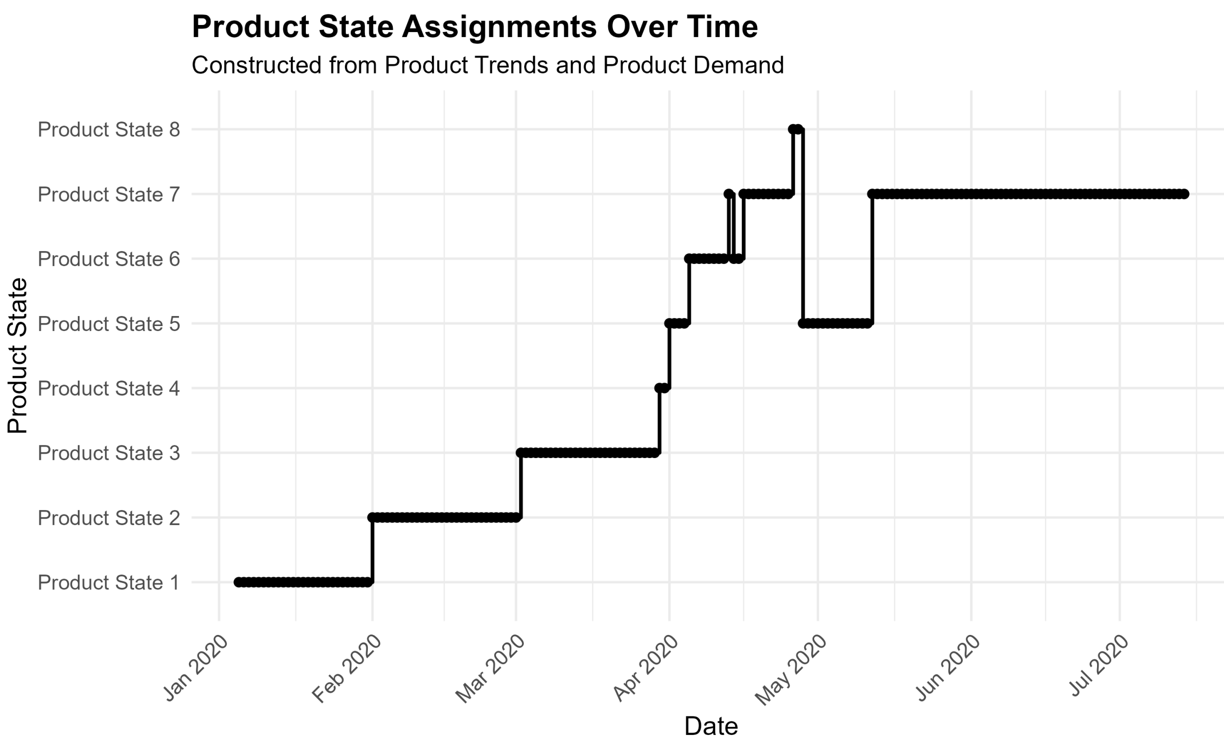


Figure 4.2: Categorical time series of product states identified from the joint assignment of product trends and product demand clusters.

As shown in Figure 4.2, transitions between product states occur throughout January through March due to changes in product trends while overall demand remains relatively stable. This indicates that, prior to the suspected COVID-19 disruptions, shifts in customer purchasing behavior were primarily driven by changes in product preferences rather than demand fluctuations. In particular, some products decreased in relative popularity while others increased over time.

During April, several transitions between product states are observed as a result of changes in both demand and product trends. In May, transitions are initially associated with changes in overall demand, followed by another transition that becomes the dominant pattern throughout the remainder of the observed period.

We summarize the identified product states in Table 4.3.

Table 4.3: Product states summary

Product State	**Count**	**Proportion (%)**	**Months Occurred**
1	27	14.1	Jan
2	30	15.6	Feb
3	28	14.6	Mar
4	2	1.0	Mar
5	18	9.4	Apr–May
6	10	5.2	Apr
7	75	39.1	Apr–Jul
8	2	1.0	Jun

As shown in Table 4.3, substantial changes in product behavior emerge from April through July. One possible explanation is the presence of inventory shortages, as approximately 20% of products (9 out of 49) recorded zero transactions during this period, accompanied by repeated shifts between demand and product trends. Additionally, labor availability may have influenced the quantity of products shipped per day, potentially contributing to new product states driven by operational capacity constraints.

We now summarize the transitions that occurred between the product states. We summarize the observed product state transitions and their corresponding behavioral changes in Table 4.4.

Table 4.4: Summary of product state transitions

Transition	**States Involved**	**Cause of Change in Behavior**
Transition 1	State 1 → State 2	Product trends
Transition 2	State 2 → State 3	Product trends
Transition 3	State 3 → State 4	Overall demand decrease
Transition 4	State 4 → State 5	Product trends
Transition 5	State 5 → State 6	Product preferences
Transition 6	State 6 → State 7	Overall demand decrease
Transition 7	State 7 → State 6	Overall demand increase
Transition 8	State 7 → State 8	Product preferences
Transition 9	State 8 → State 5	Overall demand increase
Transition 10	State 5 → State 7	Overall demand decrease

As shown in Table 4.4, product states emerge from a combination of changes in overall demand and product trends. Transitions associated with changes in product trends are suspected to reflect shifts in customer purchasing behavior that may have been influenced by inventory shortages and operational constraints.

As before we do not implement a transition matrix or n-grams as the results are very consistent suggesting overall operations and purchasing behavior were effected by COVID-19, factors outside of the businesses control.

Having identified the business states and product states, and analyzed their corresponding behaviors, we now present our conclusions regarding business performance, evaluate the effectiveness of the proposed framework, discuss unexpected findings, and provide recommendations for future analysis and operational decision-making.

Chapter 5

Conclusion

In conclusion, we were able to identify business states and product states from transactional data and from our outlined framework. However, we were unable to implement all the methods described due to the nature of the data. Regardless, the framework provided has been proven valid in understanding an overview of operational performance and in-depth purchasing trends caused by consumer preferences evaluated through product behavior and product demand.

We now conclude our study by evaluating the framework, summarizing key insights into business performance, and providing recommendations based on our findings.

5.1 Effectiveness of Framework

This study aimed to jointly capture product distributions and overall demand to evaluate changes in each component over time. Using clustering methods, we identified two product preference groups that represent general customer interest, three product demand groups that reflect transaction volume, and five product trends that capture changes in product popularity and behavior.

Product trends were modeled using the Dirichlet distribution to summarize their composition and variability. The estimated parameters indicated concentrated preferences, and evaluating these trends over time provided insight into product behavior and customer purchasing patterns.

By jointly evaluating product preferences and product demand, we constructed business states that reflected transitions in operations driven by changes in transaction volume and product composition. Further analysis combining product trends and product demand produced product states that provided more detailed insight into purchasing patterns.

The observed transitions between states indicated that, leading up to a potential inventory disruption, some products declined in relative share while others increased. This reflected changes in product trends and demand, and suggested the emergence of new customer purchasing patterns in response to fluctuations in available inventory during the initial impacts of COVID-19.

These results demonstrate that separating product probability distributions and variation in demand provides a structured approach to understanding business performance and identifying shifts in purchasing behavior over time.

We now discuss how the structure of the data limited the scope of the intended analysis.

5.1.1 Unanticipated Findings

Due to the nature of the data, not all components of the proposed framework could be applied. We outline these observations and provide context for why they occurred.

The clusters formed for product preferences and demand did not exhibit seasonal temporal transitions. In particular, transitions between clusters and between states were limited, with newer states rarely returning to earlier ones. As a result, the use of transition matrices to model state changes was not justified. Similarly, n-gram analysis was not applied because the consistency of states over extended periods limited the ability to identify meaningful sequential patterns. Additional temporal features, such as day-of-week effects, were also not evaluated due to the consistency of the clusters over time.

We also observed that one of the demand clusters exhibited degeneracy, where a wide range of transaction volumes were grouped within the same cluster across dates. In addition, overlap between certain clusters suggests that distinct demand levels were not clearly separated. This suggests that demand may not partition into clearly defined levels under the current approach. We discuss solutions to this in the Future Work section.

These findings suggest that the framework provides an overview of the data, but may require refinement to capture more subtle patterns. Despite these findings, the analysis provides insight into business performance during the study period and highlights patterns in product demand and purchasing behavior under uncertain conditions.

5.2 Insights into Business Performance

The business and product states provide a structured view of operational performance over the study period. These insights should be interpreted with caution, as the available business context is limited and additional external factors may have influenced the observed behavior.

5.2.1 Key Insights from Business States

From the business states, we observed that product demand and product preferences remained relatively stable from January through March, indicating consistent operational performance. During April and into mid-May, transitions in both demand and product preferences emerged, suggesting a period of operational disruption. In particular, several products were no longer observed in transactions, indicating potential constraints in product availability rather than shifts in customer preference.

These transitions align with the timing of COVID-19–related disruptions, including workforce limitations and supply chain constraints. Despite these changes, both demand and product preferences stabilized from mid-May through July, suggesting that the business adapted to the new operating conditions and reached a period of consistency.

We now discuss the key findings from the identified product states.

5.2.2 Key Insights from Product States

From the product states, we observed shifts in product trends during February, March, and April. Specifically, Product 286 declined while other products increased, even as overall demand remained relatively stable. This indicates changes in relative product preferences rather than overall demand.

During April through mid-May, both product trends and demand fluctuated, indicating a period of instability. These changes suggest that constraints in product availability influenced observed purchasing behavior, as only certain products were consistently available.

From mid-May through July, product trends remained stable, reinforcing the observation that customer purchasing patterns became more consistent following the initial disruption.

Based on these findings, we provide recommendations in the following section.

5.2.3 Recommendations

Operations

Based on the business states, both transaction volume and product preferences can be observed simultaneously. This provides a framework to identify periods of lower demand and adjust operational decisions accordingly, including staffing levels and inventory allocation. In addition, sustained changes in product preferences may inform decisions on product lifecycle management, such as reducing emphasis on under-performing products or identifying candidates for obsolescence.

Inventory Strategies

Based on the identified product states, inventory strategies can be informed by both demand and product trends. From January through March, demand remained relatively stable, suggesting that inventory levels could have been maintained consistently during this period. However, variation in product trends indicates that adjustments in product mix would be necessary, as certain products increased or declined in relative share despite stable overall demand.

During periods of disruption, particularly April through mid-May, fluctuations in both demand and product trends suggest that inventory strategies should be more adaptive. Monitoring changes in product trends alongside demand can help identify which products require increased stock and which may experience reduced activity.

Marketing Initiatives

Product preference clusters suggest that shifts in customer purchasing behavior may be present over time. Although seasonal effects are not explicitly confirmed, the clustering framework provides a method to identify periods in purchasing patterns. This can be used to evaluate potential seasonal trends or external influences on demand.

Product trends provide additional insight into how individual products change in relative importance over time. These trends can be used as a baseline to evaluate the effectiveness of marketing initiatives, such as promotions or pricing strategies, by assessing whether they lead to changes in transaction volume or product share.

It is also recommended that the business further investigate the decline of Product 286 prior to the observed disruption. Understanding the factors contributing to this decline may provide additional insight into customer behavior and support more informed decision-making if similar patterns arise in the future.

We now conclude the study and discuss directions for future work.

5.3 Future work

5.3.1 Business Context and Data

A limitation of our analysis was not knowing the product names and how the products relate to each other. Applying the framework to data that have more product context, as well as knowing which business the data comes from, would allow for more specific recommendations.

Within our analysis, we did not implement all of the sequence analysis methods proposed due to how consistent the sequences were. Given another dataset, we would suspect different behavior between state transitions and more seasonality patterns occurring. This could be due to the impact of COVID-19, the business is well established, or the method of how the clusters were formed.

5.3.2 Further Product Analysis

Future work may incorporate conventional product analysis methods such as ABC analysis, first-in, first-out (FIFO), and transactional cost analysis [15]. However, the scope of the present study was limited to examining how the structure of the data across different attributes influences inventory management decisions.

5.3.3 Additional Component to Analyze

During the analysis of the product demand clusters (Figure), we observed that while the clusters appear similar in magnitude, some temporal variation may not be fully captured when viewed in isolation. In particular, fluctuations in demand between observations are not explicitly represented within the cluster assignments.

By examining the trajectory as correlation of demand across time within clusters, these fluctuations can be recovered, allowing for the identification of periods of increasing or decreasing demand. This provides additional context for interpreting the clusters and improves understanding of how demand evolves over time.

5.3.4 Dynamic Tree Cut

A conservative approach was used to determine the number of clusters by applying the silhouette method, which effectively selects a fixed number of clusters and corresponds to a static cut of the hierarchical clustering dendrogram. While this ensures well-separated and stable clusters, it imposes a global structure on the data.

During the analysis of product demand, there were indications that one partition contained a larger variance than the others. However, increasing the total number of clusters to capture potential subclusters would have forced splits across other clusters, potentially reducing their interpretability and coherence.

Dynamic tree cutting offers a more flexible alternative by adaptively identifying clusters based on the local structure of the dendrogram. This allows subclusters within larger groups to be detected without requiring a global increase in the number of clusters.

However, this method introduces additional complexity, as it requires careful tuning of parameters and may lead to over-partitioning if not properly constrained. Despite this, dynamic tree cutting may be better suited for uncovering subtle structured in the data when nuances are important.

Bibliography

[1] Jukka Corander, Ulpu Remes, and Timo Koski. “On the Jensen–Shannon divergence and the variation distance for categorical probability distributions”. In: *Kybernetika* 57.6 (2021), pp. 879–907. DOI: 10.14736/kyb-2021-6-0879. URL: https://www.kybernetika.cz/content/2021/6/879/paper.pdf.

[2] Emily B. Fox et al. “An HDP-HMM for Systems with State Persistence”. In: *Proceedings of the 25th International Conference on Machine Learning (ICML)*. Helsinki, Finland: Omnipress, 2008, pp. 312–319. URL: https://ics.uci.edu/~sudderth/papers/icml08.pdf.

[3] R. E. van Ruitenbeek, G. M. Koole, and S. Bhulai. “A hierarchical agglomerative clustering for product sales forecasting”. In: *Decision Analytics Journal* 8 (2023), p. 100318. DOI: 10.1016/j.dajour.2023.100318. URL: https://www.sciencedirect.com/science/article/pii/S2772662223001583.

[4] Benedetta Baldi et al. “Consumer-Centric Supply Chain Management: A Literature Review, Framework, and Research Agenda”. In: *Journal of Business Logistics* 45.4 (2024). e12399 JBL-Aug-2023-8614.R3, e12399. DOI: 10.1111/jbl.12399. URL: https://onlinelibrary.wiley.com/doi/10.1111/jbl.12399.

[5] S. Dinesh Kumar and A. M. Santhosh. “The impact of inventory management on business performance: A comprehensive study”. In: *International Journal of Research in Fi-*

nance and Management 7.2 (2024), pp. 592–595. URL: https://www.allfinancejournal.com/archives/2024/vol7issue2/PartF/7-2-88-152.pdf.

[6] Jiayu Lin. "On the Dirichlet Distribution". Master of Science thesis. Kingston, Ontario, Canada: Queens University, Sept. 2016. URL: https://mast.queensu.ca/~communications/Papers/msc-jiayu-lin.pdf.

[7] Brian S. Everitt et al. *Cluster Analysis*. 5th ed. Chichester, UK: Wiley, 2011. ISBN: 9780470749913.

[8] Peter J. Rousseeuw. "Silhouettes: A graphical aid to the interpretation and validation of cluster analysis". In: *Journal of Computational and Applied Mathematics* 20 (1987), pp. 53–65. DOI: 10.1016/0377-0427(87)90125-7.

[9] Hugues Salamin and Alessandro Vinciarelli. "Introduction to Sequence Analysis for Human Behavior Understanding". In: *Social Signal Processing*. Ed. by Alessandro Vinciarelli, Maja Pantic, and Hervé Bourlard. Cambridge University Press, 2011, pp. 165–185.

[10] Christopher D. Manning and Hinrich Schütze. *Foundations of Statistical Natural Language Processing*. Cambridge, MA: MIT Press, 1999. ISBN: 9780262133609. URL: https://icog-labs.com/wp-content/uploads/2014/07/Christopher_D._Manning_Hinrich_Sch%C3%BCtze_Foundations_Of_Statistical_Natural_Language_Processing.pdf.

[11] M. Kechinov. *eCommerce Purchase History from Electronics Store*. Collected by the Open CDP Project (https://rees46.com/en/open-cdp). Accessed: 2026-02-23. 2020. URL: https://www.kaggle.com/datasets/mkechinov/ecommerce-purchase-history-from-electronics-store.

[12] United Nations Conference on Trade and Development (UNCTAD). *Shipping Data: UNCTAD Releases New Seaborne Trade Statistics*. Published April 23, 2025. Accessed: 2026-05-03. 2025. URL: https://unctad.org/news/shipping-data-unctad-releases-new-seaborne-trade-statistics.

[13] United States International Trade Commission. *Special Topic*. Accessed: 2026-05-02. 2020. URL: https://www.usitc.gov/research_and_analysis/tradeshifts/2020/special_topic.html.

[14] Sean M. Smith and Roxanna Edwards. "Unemployment rises in 2020, as the country battles the COVID-19 pandemic". In: *Monthly Labor Review* (June 2021). Accessed: 2026-05-02. URL: https://www.bls.gov/opub/mlr/2021/article/unemployment-rises-in-2020-as-the-country-battles-the-covid-19-pandemic.htm.

[15] Mwondha Kennedy. "Effect of Inventory Management on Sales Performance: A Case Study of National Medical Stores". Bachelor of Science in Procurement and Supply Chain Management. Undergraduate thesis. Mbarara University of Science and Technology, Feb. 2019. URL: https://www.academia.edu/40509401/EFFECT_OF_INVENTORY_MANAGEMENT_ON_SALES_PERFORMANCE_A_CASE_STUDY_OF_NATIONAL_MEDICAL_STORES.

Appendix A

Modeled Dirichlet Parameters and Expected Values

Table A.1: Estimated Dirichlet concentration parameters across product trends.

Product	Trend 1	Trend 2	Trend 3	Trend 4	Trend 5
1	499.59	411.32	522.01	178.69	75.51
2	98.09	81.41	102.82	32.21	13.60
3	397.26	297.52	341.34	116.12	51.37
4	168.72	128.54	125.11	33.08	12.15
5	253.44	227.73	218.45	61.92	25.46
6	130.90	109.90	112.46	32.29	12.26
7	15.87	13.38	13.47	3.76	1.53
9	153.79	131.57	128.73	32.98	13.78
10	149.87	129.00	131.06	39.50	13.34
11	25.88	19.76	22.13	6.09	2.12

Note: Values represent fitted α parameters from the Dirichlet models.

Product	Trend 1	Trend 2	Trend 3	Trend 4	Trend 5
12	53.68	42.45	41.74	12.87	4.67
13	29.99	25.66	25.83	6.29	3.17
14	5.90	18.77	4.81	2.00	0.57
15	40.35	37.85	34.63	9.85	3.25
16	44.53	39.45	31.19	7.68	3.10
17	46.24	39.17	32.31	8.98	2.48
20	38.36	34.84	33.45	9.58	2.86
23	14.95	11.43	10.43	2.93	1.13
24	14.19	11.75	11.61	3.21	1.41
25	12.30	9.91	9.83	3.15	0.86
26	32.74	25.03	18.73	5.58	1.87
32	15.38	11.55	10.91	3.30	0.69
54	44.13	38.95	36.37	9.44	2.88
55	13.48	11.15	11.10	3.38	1.04
56	19.91	19.34	15.44	4.36	0.88
58	11.61	11.42	10.38	3.37	0.81
62	16.81	15.51	15.47	4.19	1.41
63	16.74	14.14	10.91	3.16	1.23
74	16.05	14.62	13.38	3.92	1.04
286	941.50	696.98	662.71	104.69	0.62
305	56.93	45.51	40.89	9.74	0.48
308	81.19	58.28	24.67	6.18	0.46
311	7.68	7.68	20.23	2.01	0.42

Note: Values represent fitted α parameters from the Dirichlet models.

Product	**Trend 1**	**Trend 2**	**Trend 3**	**Trend 4**	**Trend 5**
313	12.84	17.43	11.74	2.62	0.45
420	47.22	31.37	30.12	6.17	0.48
527	22.43	16.37	13.22	3.11	0.42
561	55.25	40.28	37.87	5.11	0.44
628	20.40	15.54	12.36	2.86	0.53
668	18.47	12.72	8.15	2.66	0.43
792	30.98	24.35	27.98	4.56	0.87
1021	15.13	11.14	10.52	3.66	0.57
1086	16.07	12.77	13.68	4.75	0.62
1283	10.39	19.54	15.34	4.01	0.44
1347	14.06	12.98	11.89	5.82	1.36
1368	13.83	8.55	11.79	4.55	0.93
1438	25.31	13.83	14.24	4.37	0.77
1473	36.16	17.76	26.30	4.35	1.69
1478	26.63	23.95	17.55	5.11	1.37
Others	903.20	782.00	770.97	197.77	97.43

Note: Values represent fitted α parameters from the Dirichlet models.

Table A.2: Mean product proportions across product trends (rounded to one decimal place).

Product	**Trend 1**	**Trend 2**	**Trend 3**	**Trend 4**	**Trend 5**
1	10.9%	11.0%	13.9%	18.3%	21.1%
2	2.1%	2.1%	2.7%	3.2%	3.7%
3	8.6%	7.9%	9.0%	11.8%	14.1%
4	3.6%	3.4%	3.3%	3.3%	3.3%
5	5.5%	6.1%	5.8%	6.3%	7.0%
6	2.8%	2.9%	2.9%	3.2%	3.3%
7	0.3%	0.3%	0.3%	0.3%	0.4%
9	3.3%	3.5%	3.4%	3.3%	3.8%
10	3.2%	3.4%	3.4%	3.9%	3.7%
11	0.5%	0.5%	0.5%	0.5%	0.6%
12	1.1%	1.1%	1.0%	1.3%	1.3%
13	0.6%	0.6%	0.6%	0.6%	0.9%
14	0.0%	0.5%	0.1%	0.1%	0.1%
15	0.8%	0.9%	0.9%	0.9%	0.9%
16	0.9%	1.0%	0.8%	0.7%	0.8%
17	0.9%	1.0%	0.8%	0.8%	0.7%
20	0.8%	0.9%	0.8%	0.9%	0.7%
23	0.2%	0.2%	0.2%	0.2%	0.3%
24	0.2%	0.2%	0.2%	0.2%	0.4%
25	0.2%	0.2%	0.2%	0.2%	0.2%
26	0.6%	0.6%	0.4%	0.5%	0.5%

Note: Values represent expected proportions $E[X_j] = \alpha_j / \sum_r \alpha_r$.

Product	Trend 1	Trend 2	Trend 3	Trend 4	Trend 5
32	0.3%	0.2%	0.2%	0.2%	0.1%
54	0.9%	1.0%	0.9%	0.8%	0.8%
55	0.2%	0.2%	0.2%	0.2%	0.3%
56	0.4%	0.4%	0.3%	0.3%	0.2%
58	0.2%	0.2%	0.2%	0.2%	0.2%
62	0.3%	0.3%	0.3%	0.3%	0.4%
63	0.3%	0.3%	0.2%	0.2%	0.3%
74	0.3%	0.3%	0.3%	0.3%	0.2%
286	20.6%	18.7%	17.7%	10.8%	0.4%
305	1.2%	1.2%	1.0%	0.9%	0.0%
308	1.7%	1.5%	0.6%	0.6%	0.0%
311	0.1%	0.1%	0.5%	0.1%	0.0%
313	0.3%	0.4%	0.3%	0.2%	0.0%
420	1.0%	0.8%	0.7%	0.5%	0.0%
527	0.4%	0.4%	0.3%	0.2%	0.0%
561	1.1%	1.0%	0.9%	0.4%	0.0%
628	0.4%	0.3%	0.3%	0.2%	0.1%
668	0.3%	0.3%	0.2%	0.1%	0.0%
792	0.6%	0.6%	0.7%	0.4%	0.2%
1021	0.3%	0.2%	0.2%	0.3%	0.1%
1086	0.3%	0.3%	0.3%	0.4%	0.1%
1283	0.2%	0.5%	0.3%	0.3%	0.0%
1347	0.2%	0.3%	0.2%	0.5%	0.4%

Note: Values represent expected proportions $E[X_j] = \alpha_j / \sum_r \alpha_r$.

Product	**Trend 1**	**Trend 2**	**Trend 3**	**Trend 4**	**Trend 5**
1368	0.2%	0.2%	0.3%	0.4%	0.2%
1438	0.5%	0.3%	0.3%	0.3%	0.2%
1473	0.7%	0.4%	0.6%	0.3%	0.5%
1478	0.5%	0.7%	0.4%	0.4%	0.4%
Others	19.7%	20.9%	20.5%	20.2%	27.3%

Note: Values represent expected proportions $E[X_j] = \alpha_j / \sum_r \alpha_r$.